

数据驱动前沿算法与发展趋势

丁文超





大纲



▣ 端到端自动驾驶

▣ 自动驾驶与多模态大模型

深蓝学院



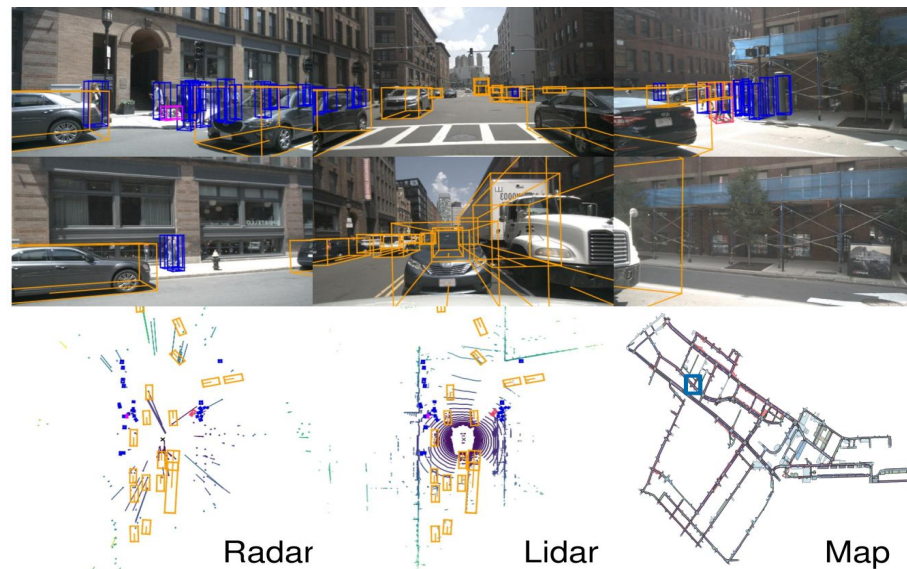
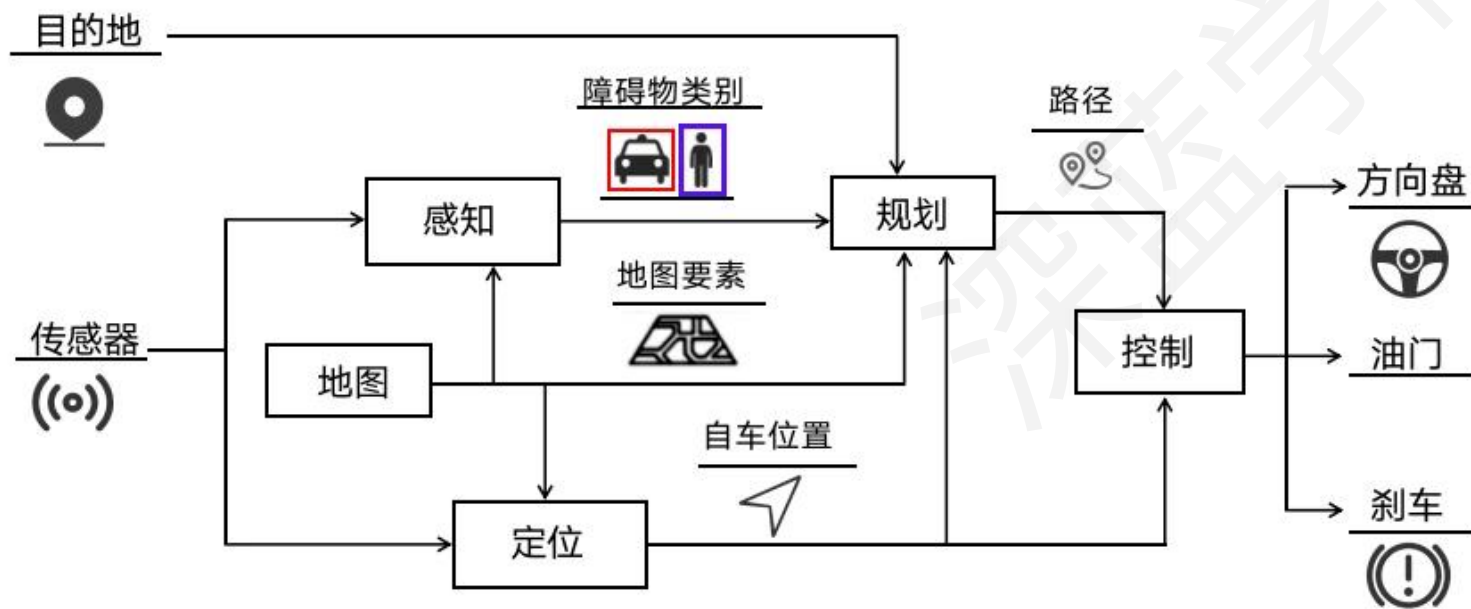


端到端自动驾驶



模块化系统与端到端系统

- 模块化自动驾驶系统
 - 优势：可维护性和可拓展性高；可解释性强；安全性和可靠性强
 - 劣势：感知模块结果通过有限种类语义的点、线、多边形对现实世界进行抽象，成为系统天花板



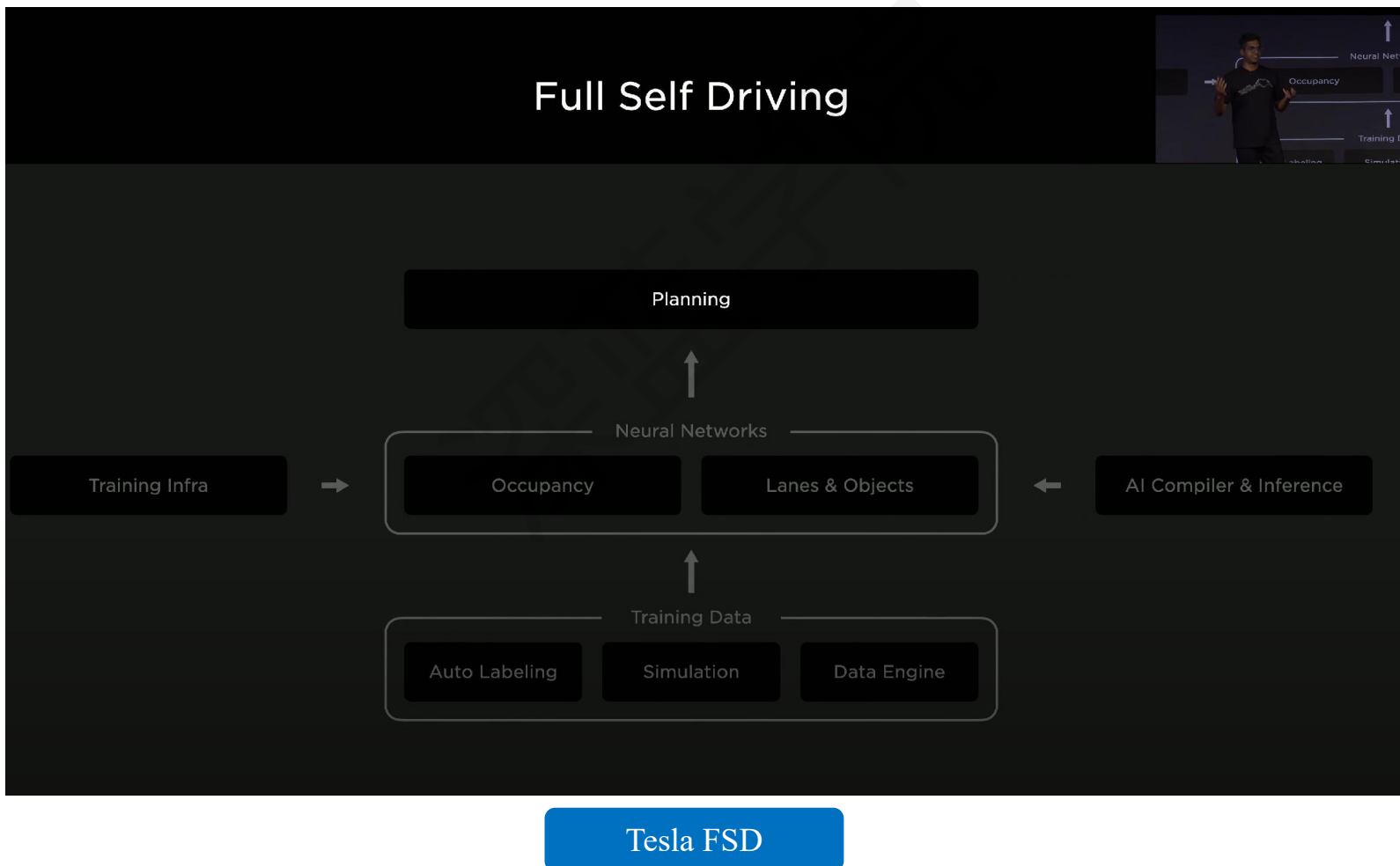


端到端自动驾驶



□ 模块化系统与端到端系统

- 端到端自动驾驶系统
 - 本质：全量感知信息到决策规划的无损传输



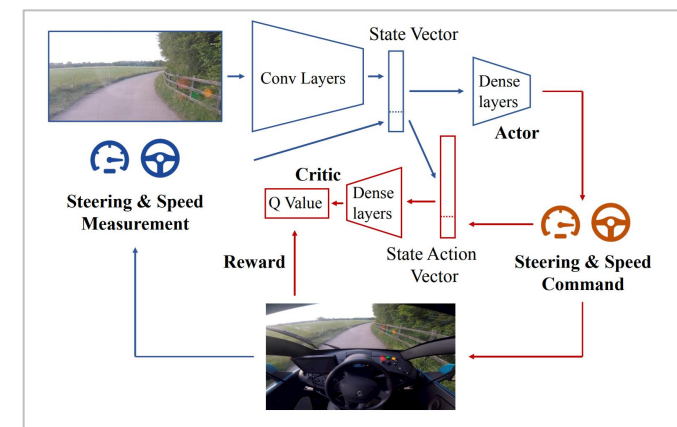
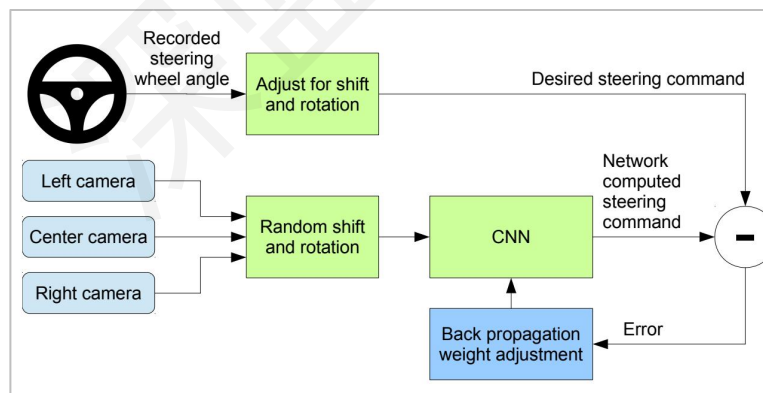
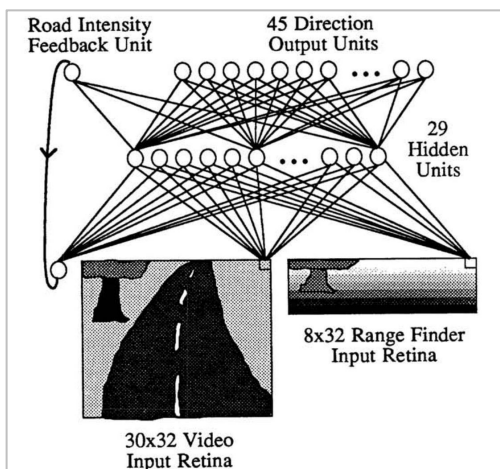
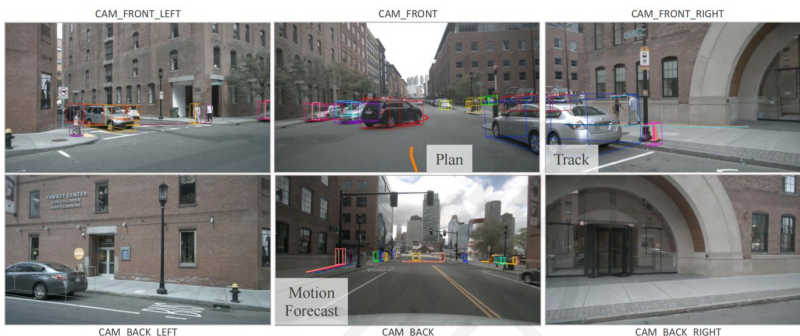


端到端自动驾驶



发展脉络与趋势

- 端到端自动驾驶早期尝试
- 基于模仿学习的端到端系统
- 基于强化学习的端到端系统
- 架构升级与多模态融合
- 模块化端到端系统
- 端到端系统评测



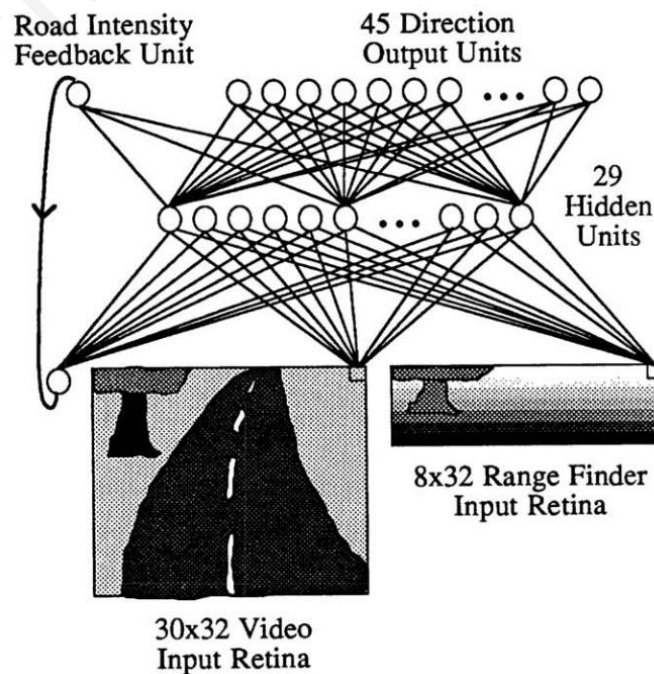


端到端自动驾驶早期尝试



▣ ALVINN

- 1988年CMU Autonomous Land Vehicle In a Neural Network
- 输入：相机拍摄图片；激光测距图片
- 输出：方向盘转角
- 网络：3层MLP
- 道路强度反馈单元：表征道路相对非道路的亮暗情况



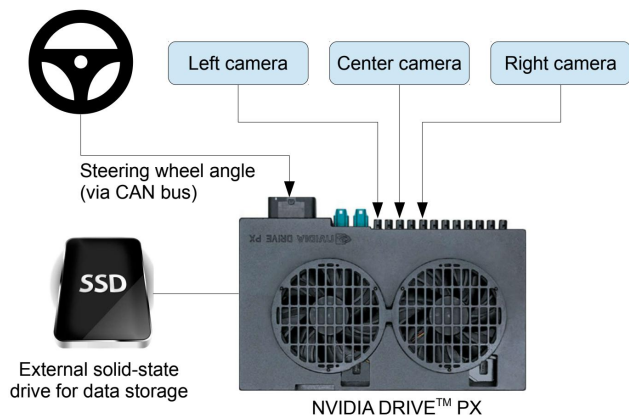


基于模仿学习的端到端系统

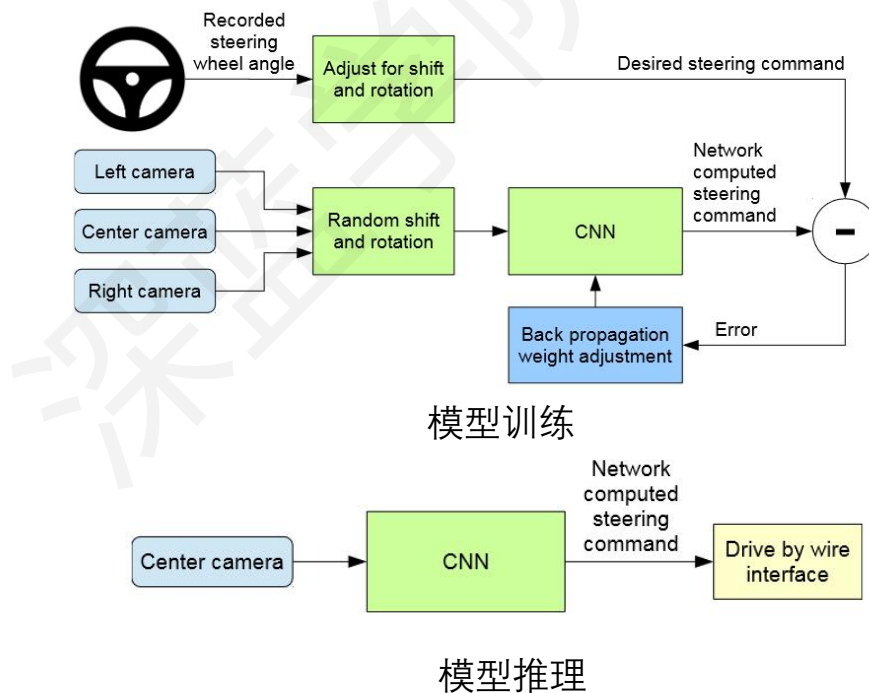


□ NVIDIA-E2E

- 首次提出观点：与显式模块化系统相比，端到端系统同时优化所有处理步骤，获取更好的性能以及更小的系统。
- 输入：视觉感知结果
- 输出： $\frac{1}{r}$ ， r 为转向半径。使用 r 使系统独立于汽车几何形状；使用倒数以处理直线行驶时的奇点。
- 网络：卷积神经网络

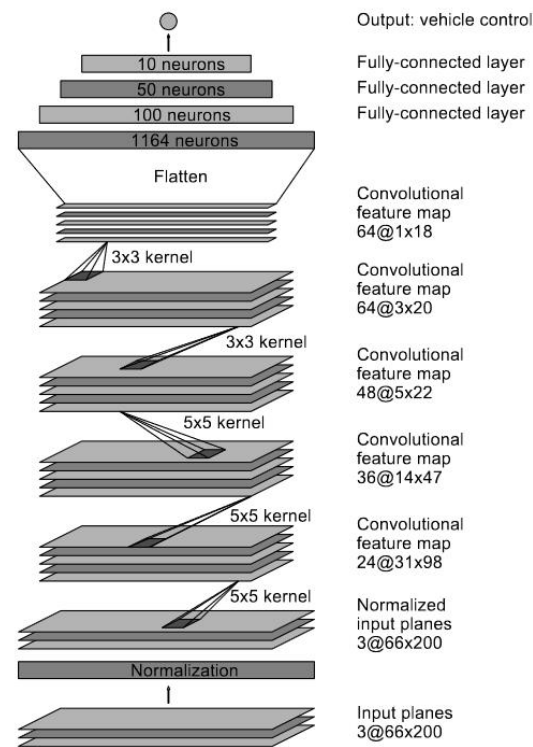


硬件平台



模型训练

模型推理



网络结构

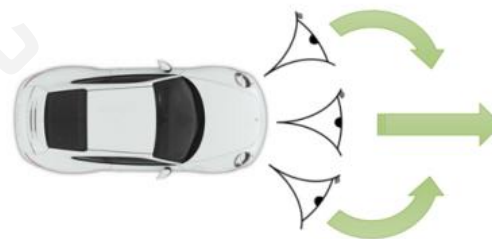




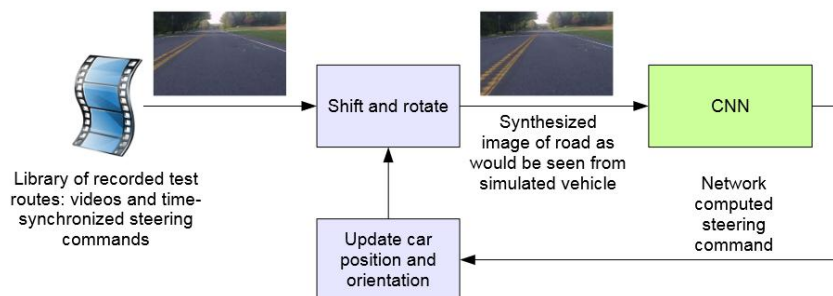
基于模仿学习的端到端系统

□ NVIDIA-E2E

- 数据增强以处理OOD(Out of Distribution)问题
 - 所采集数据只包含在道路中心线上行驶的完美数据。需生成从道路边缘驶向道路中心线的纠错数据。
 - 正样本：前视相机采集数据
 - 标签：行驶轨迹的转向半径
 - 负样本：带偏角数据（源自 偏置相机 或 施加旋转的前视相机）
 - 标签：2s内能够驶回道路中心线的转向半径
- 路测前仿真测试性能
 - 以人类驾驶录像为数据源，循环进行：预测转向、模拟车辆位置和朝向、模拟器修改对应帧视角。
 - 以自动驾驶的时长占比作为评估指标。每次人工接管时长默认为6s。



$$\text{autonomy} = \left(1 - \frac{(\text{number of interventions}) \cdot 6 \text{ seconds}}{\text{elapsed time [seconds]}}\right) \cdot 100$$



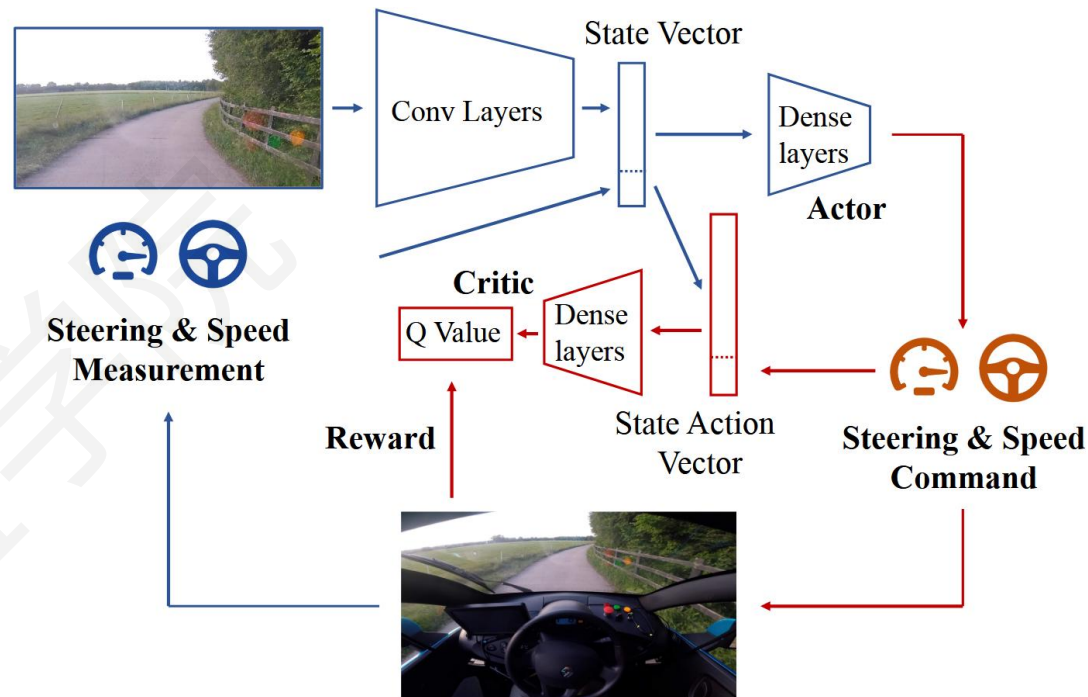


基于强化学习的端到端系统



□ 作为马尔可夫决策过程（MDP）的驾驶

- 状态集合 S ：使用单目摄像机图像以及观察到的车速和转向角作为状态的近似。原始图像通过卷积网络直接输入强化学习算法，或者使用变分自编码器(VAE)进行图像压缩表示。
- 动作集合 A ：油门、刹车、转向。
- 奖励函数 R ：将奖励定义为前进速度，并在违反交通规则时结束一个episode——因此给定状态 $V(s_t)$ 的值对应于行驶的平均距离。





基于强化学习的端到端系统

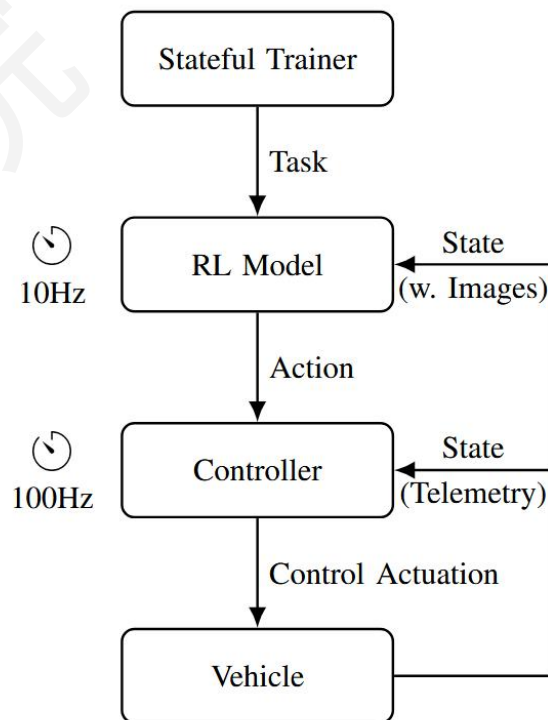


基于任务的训练架构

- 在真实世界环境中运行的全尺寸机器人车辆上部署强化学习算法需要调整通用训练程序，以考虑驾驶员干预和影响训练的外部变量。

```
1: while True do
2:   Request task
3:   Waiting for environment reset
4:   if task is train then
5:     Run episode with noisy policy
6:     if exploration time is over then
7:       Optimise model
8:     end if
9:   else if task is test then
10:    Run episode with optimal policy
11:   else if task is undo then
12:    Revert previous train/test task
13:   else if task is done then
14:    Exit experiment
15:   end if
16: end while
```

(a) Task-based workflow for on-vehicle training



(b) Policy execution architecture, used to run episodes during model training or testing.



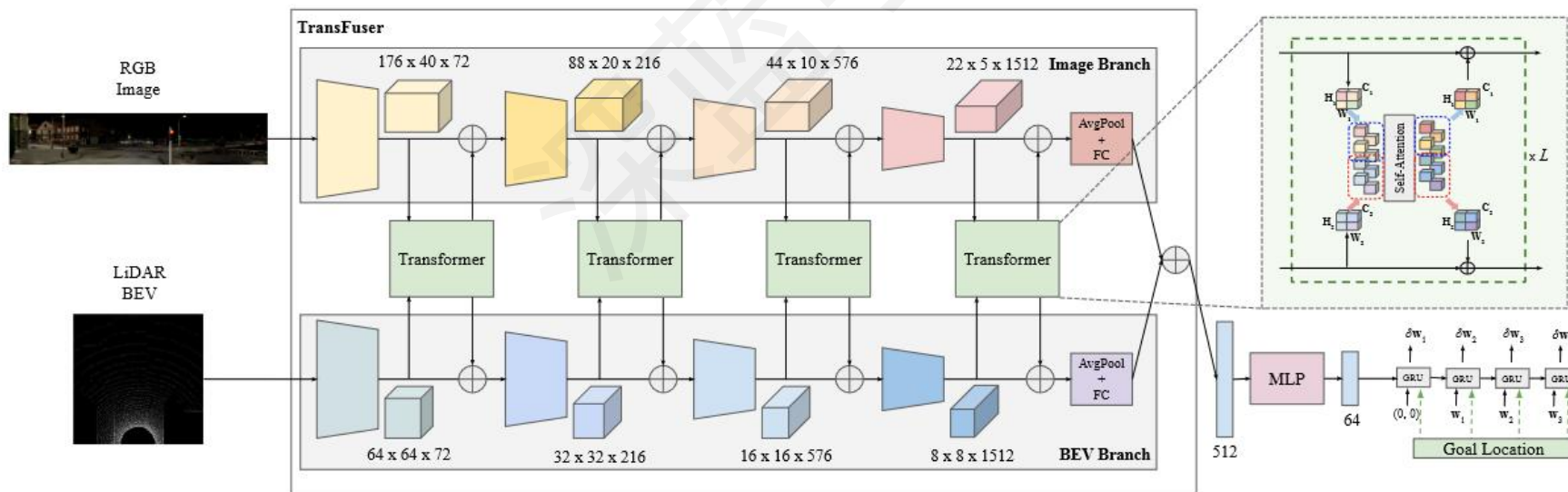


架构升级与多模态融合



TransFuser

- 基于注意力机制，多分辨率下融合多模态数据表征
- 输入：
 - 激光雷达点云：转变为 $256 \times 256 \times 2$ 的伪图像，最后一个维度代表在地平面上方/下方的点；
 - 图像：三个环视图像，剪裁为 704×160 ，FOV为 132° ；
- 输出：
 - 以自车为中心， $T = 4s$ 内的路径点。





架构升级与多模态融合



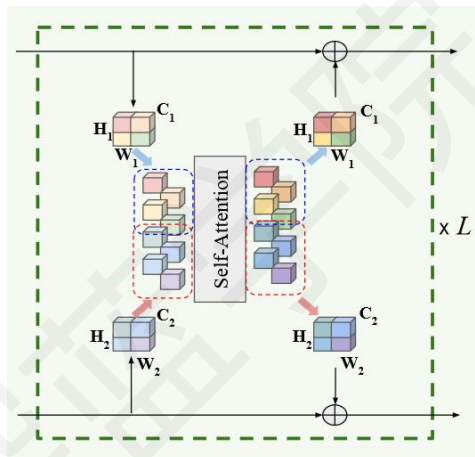
TransFuser

- 模态融合
 - 来自两个模态的特征视作等价的特征向量，通过self-attention进行特征融合

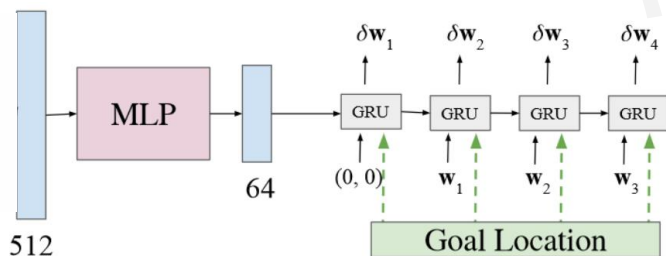
$$\mathbf{Q} = \mathbf{F}^{in} \mathbf{M}^q, \mathbf{K} = \mathbf{F}^{in} \mathbf{M}^k, \mathbf{V} = \mathbf{F}^{in} \mathbf{M}^v$$

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{D_k}}\right)\mathbf{V}$$

$$\mathbf{F}^{out} = \text{MLP}(\mathbf{A}) + \mathbf{F}^{in}$$



- 路径规划
 - 融合后的特征降维后，结合当前位置、目标位置，使用GRU预测下一个路点。使用专家轨迹进行监督。



$$\mathcal{L} = \sum_{t=1}^T \|\mathbf{w}_t - \mathbf{w}_t^{gt}\|_1$$

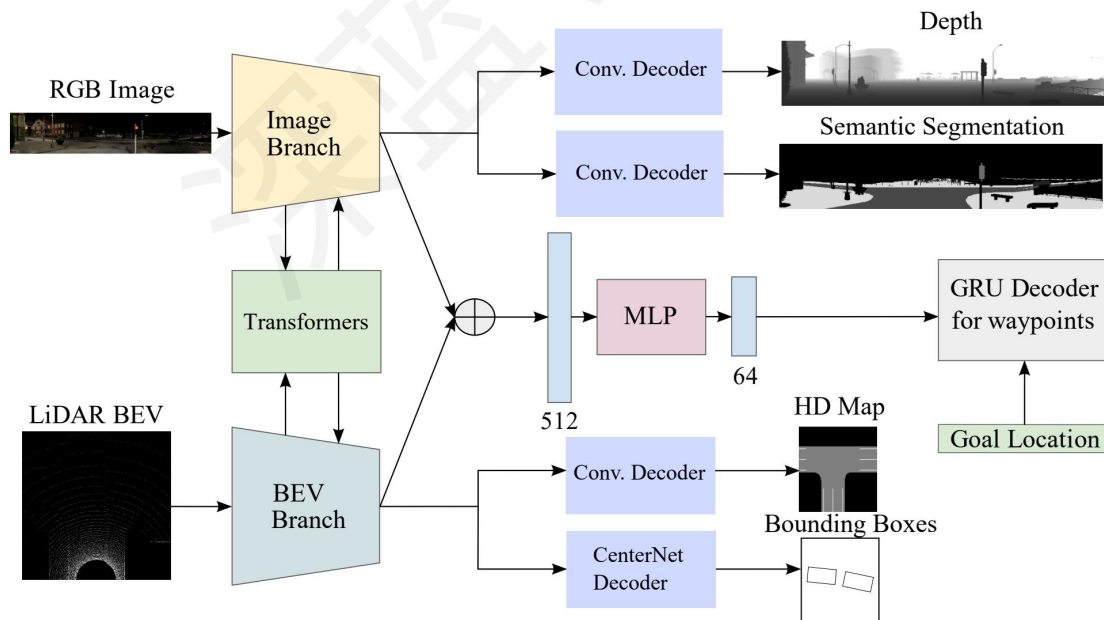




架构升级与多模态融合

TransFuser

- 辅助任务
 - 图像分支
 - 深度估计
 - 语义分割: (1) unlabeled, (2) vehicle, (3) road, (4) red light, (5) pedestrian, (6) lane marking, (7) sidewalk
 - 点云分支
 - 高精地图预测: (1) road, (2) lane marking, (3) other
 - 车辆检测



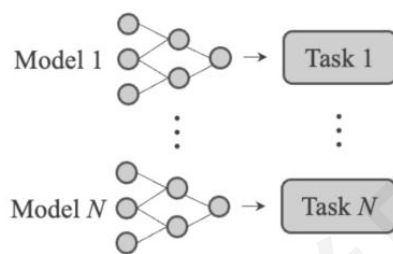


模块化端到端系统

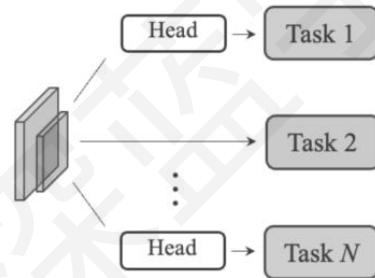


UniAD

- 聚焦规划任务端到端联合优化网络
- 模块化实现提高可解释性与安全性
- 避免级联误差，提升前序模块效果



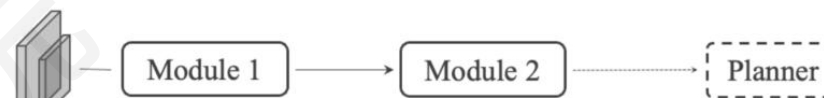
(a) Standalone Models



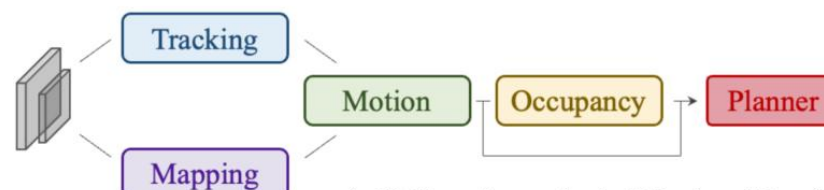
(b) Multi-task Framework



(c.1) Vanilla Solution



(c.2) Explicit Design



(c.3) Planning-oriented Design (Ours)

perception — prediction — planning —

(c) End-to-end Autonomous Driving



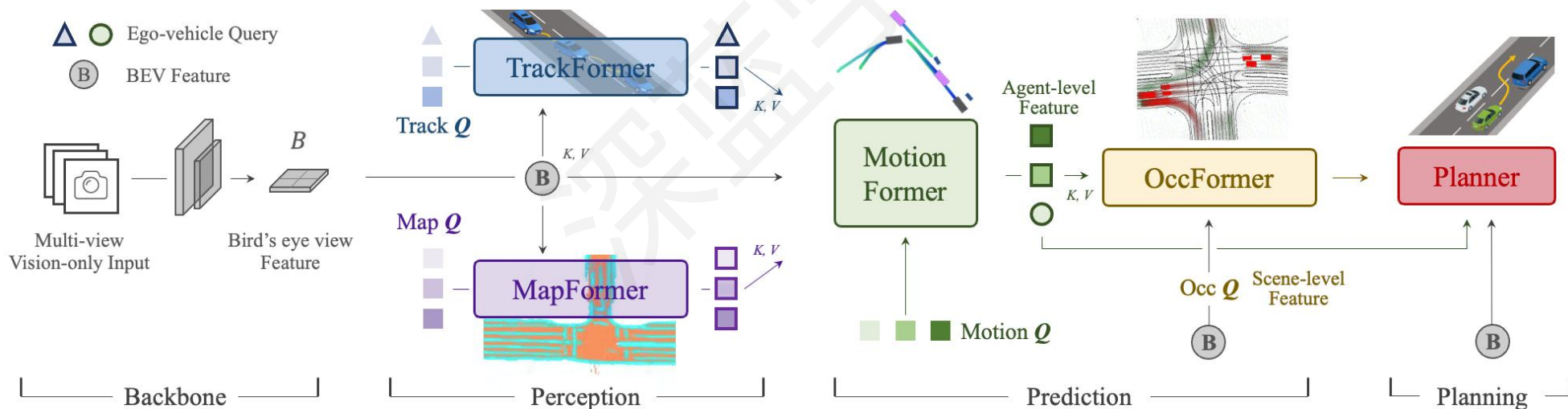


模块化端到端系统



UniAD

- 以安全规划为目标的模块化架构设计
 - 感知-目标检测与跟踪：物体行为建模
 - 感知-在线建图：车道线引导驾驶
 - 预测-运动预测：与环境动态物体交互
 - 预测-占据流预测：可行驶区域建模



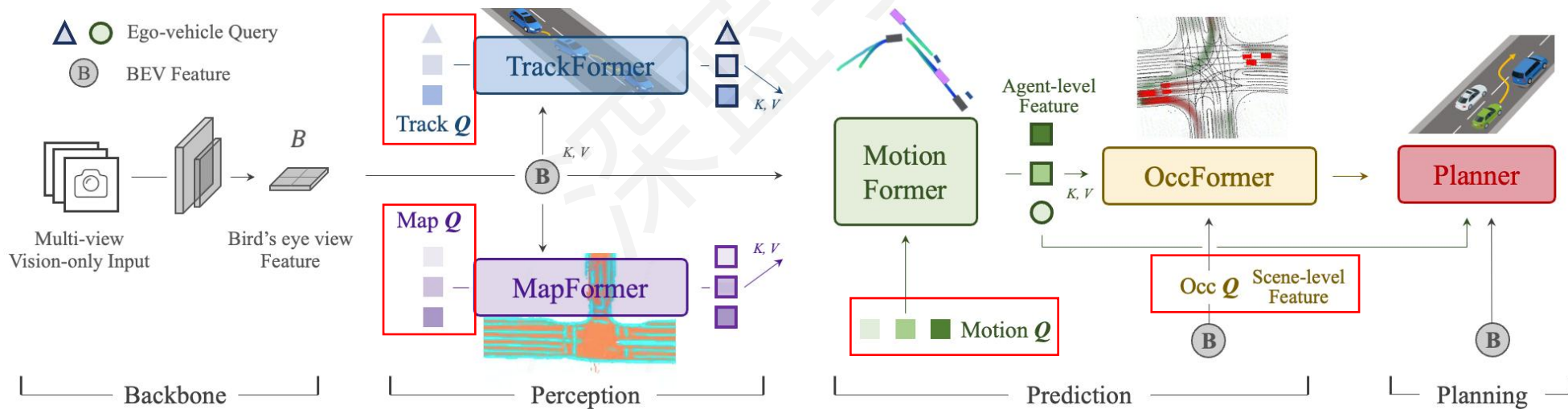


模块化端到端系统



UniAD

- 使用统一的Transformer架构，通过Query实现信息端到端传输
 - Track Q: 建模每个agent
 - Map Q: 建模每个地图实例（车道线、人行横道等）
 - Motion Q: 建模每条预测轨迹模式
 - Occ Q: 场景级别特征，建模每个时间步的场景信息



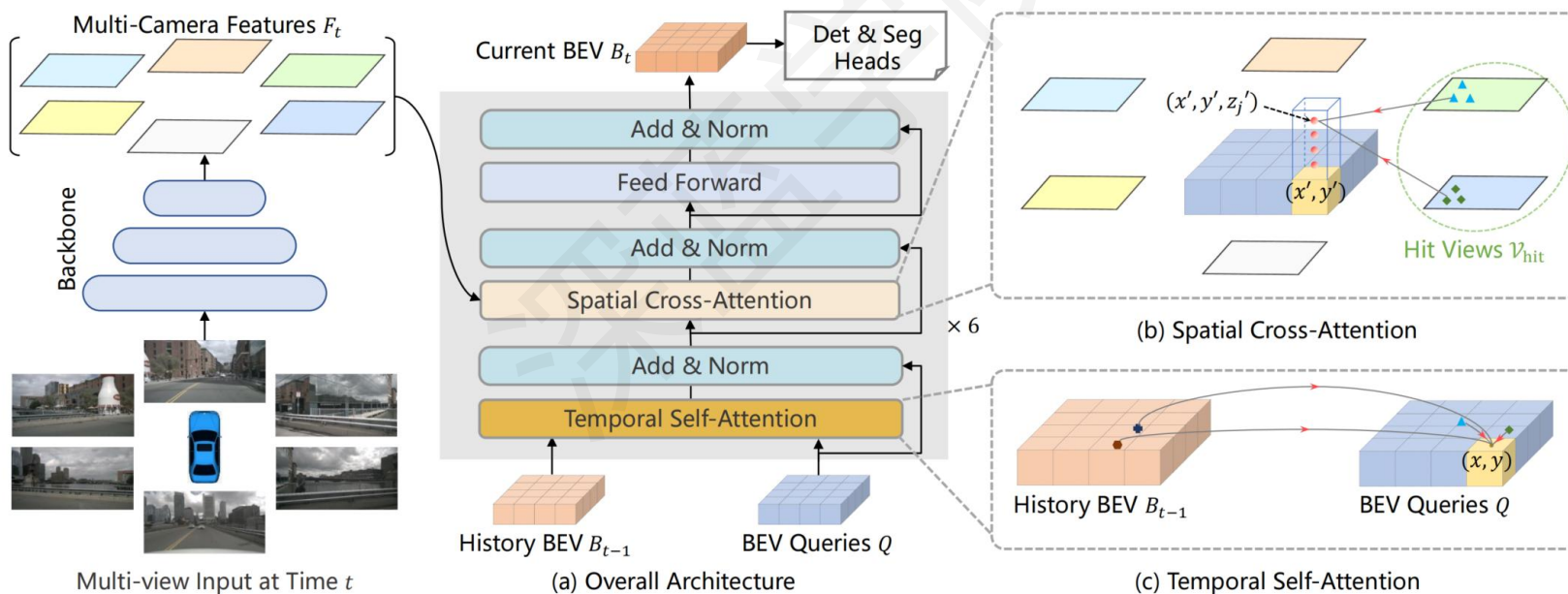


模块化端到端系统

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Backbone

- 使用BEVFormer提取多视角图片特征到BEV下
- t 时刻的BEV Queries Q 与 $t-1$ 时刻的BEV特征 B_{t-1} 通过attention机制进行时序融合，得到 B_t



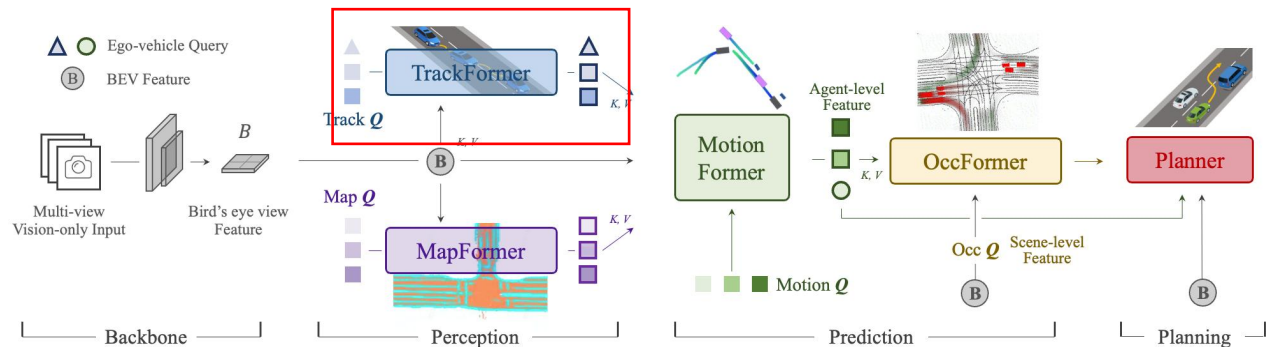


模块化端到端系统

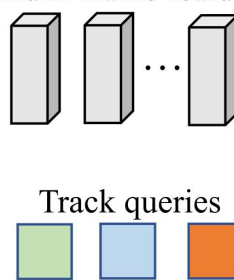
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TrackFormer

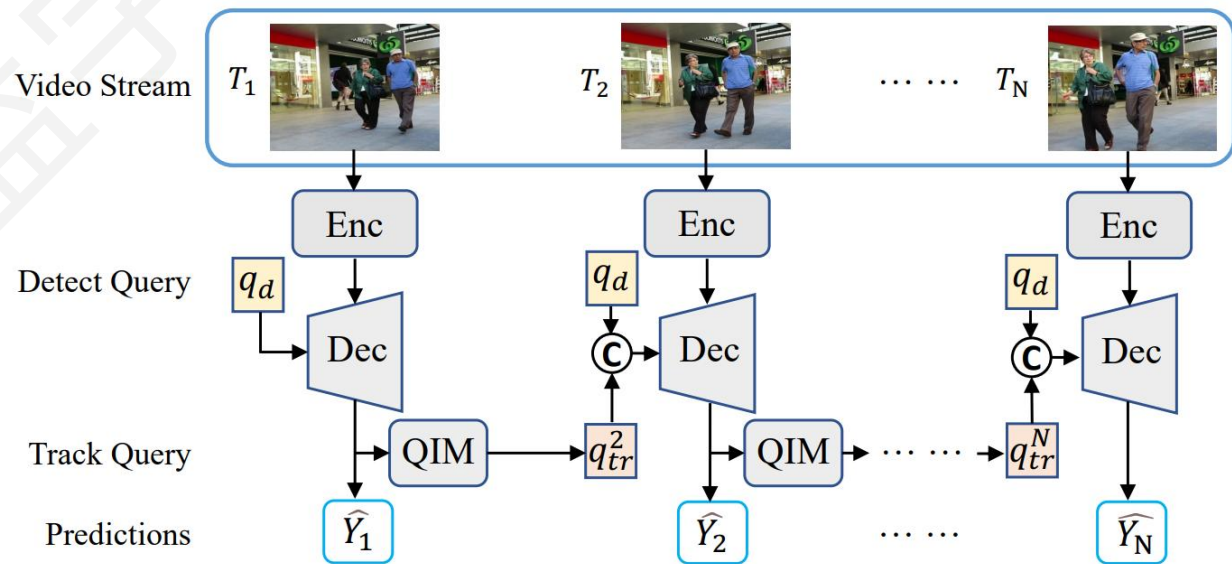
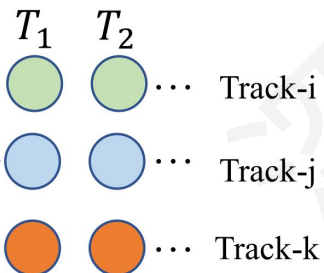
- 提取场景动态agent特征，并保持帧间一致性
- 继承backbone使用的bevformer的检测query，参考MOTR引入QIM（query interaction module）
- Query在每帧自主更新并追踪物体



Multi-frame features



Iterative Update

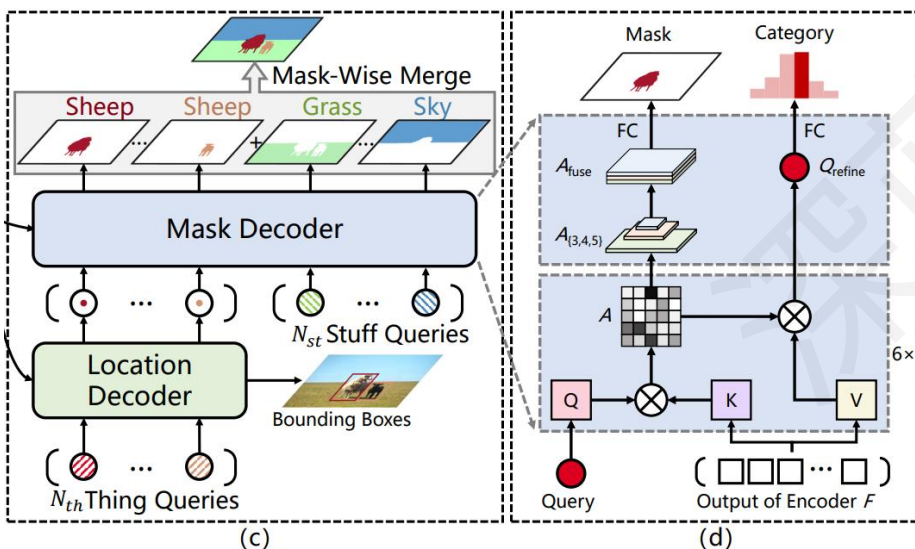
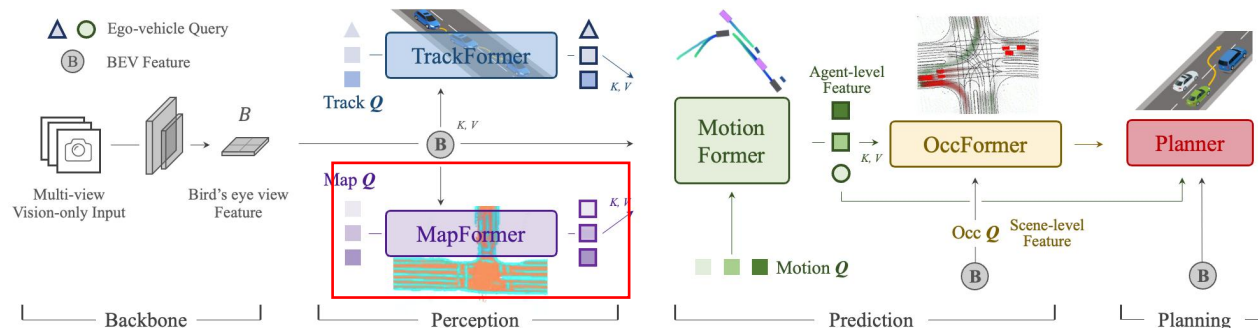




模块化端到端系统

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- MapFormer
 - 提取场景静态地图元素实例级特征
 - 使用Panoptic SegFormer对BEV栅格图进行全景分割
 - 将lane、divider、crossing作为thing，将drivable area作为stuff



Algorithm 1: Mask-Wise Merging

```
def MaskWiseMergeing(c, s, m):
    # category  $c \in \mathbb{R}^N$ 
    # confidence score  $s \in \mathbb{R}^N$ 
    # mask  $m \in \mathbb{R}^{N \times H \times W}$ 
    SemMsk = np.zeros(H, W)
    IdMsk = np.zeros(H, W)
    order = np.argsort(-s)
    id = 0
    for i in order:
         $m_i = m[i] > 0.5$  & ( $\text{SemMsk} == 0$ )
        if  $s[i] < \tau_{\text{cnf}}$  or  $\frac{m_i}{m[i] > 0.5} < \tau_{\text{keep}}$ :
            continue
        SemMsk[ $m_i$ ] =  $c[i]$ 
        IdMsk[ $m_i$ ] = id
        id += 1
    return SemMsk, IdMsk
```





模块化端到端系统

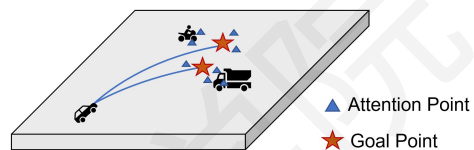
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MotionFormer

- 通过三个并列的attention模块，引入 Agent-Agent、Agent-Map、Agent-Goal交互

$$Q_{a/m} = \text{MHCA}(\text{MHSA}(Q), Q_A/Q_M)$$

$$Q_g = \text{DeformAttn}(Q, \hat{\mathbf{x}}_T^{l-1}, B)$$



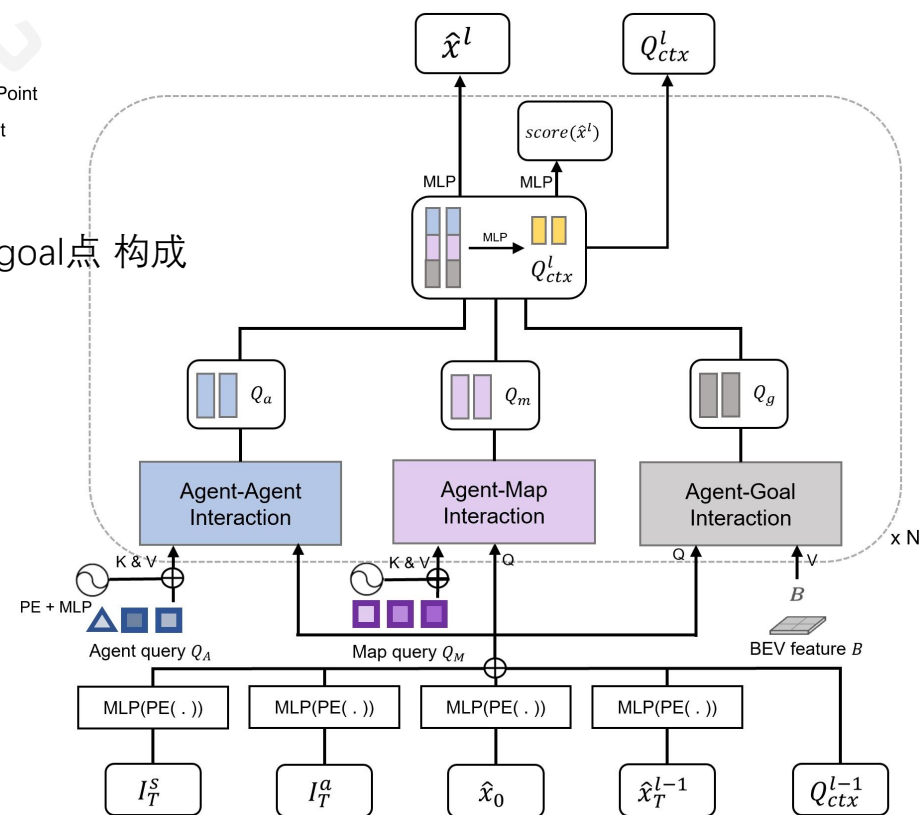
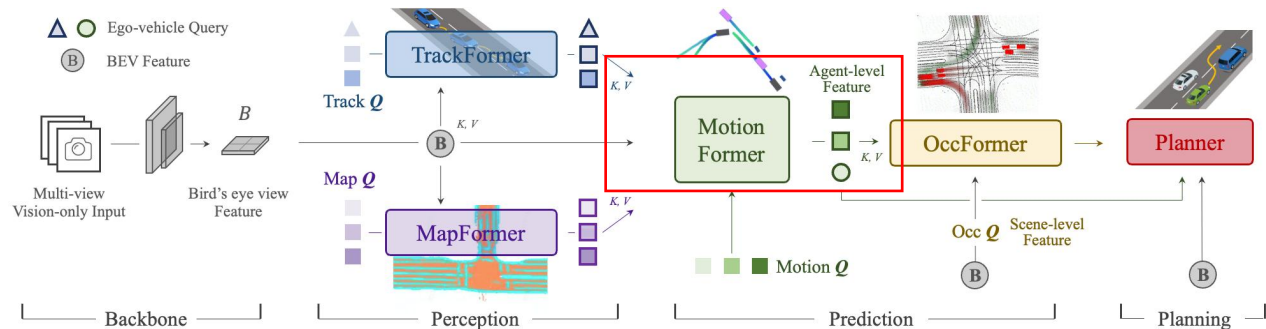
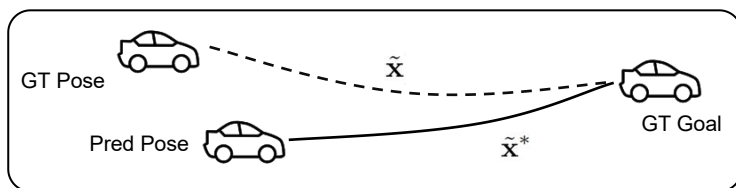
- Motion query由scene-level anchor、agent-level anchor、当前位置、预测goal点 构成

$$Q_{\text{pos}} = \text{MLP}(\text{PE}(I^s)) + \text{MLP}(\text{PE}(I^a)) + \text{MLP}(\text{PE}(\hat{\mathbf{x}}_0)) + \text{MLP}(\text{PE}(\hat{\mathbf{x}}_T^{l-1}))$$

- 结合目标点、动力学约束，非线性优化调整目标轨迹，解决上游误差

$$\tilde{\mathbf{x}}^* = \arg \min_{\tilde{\mathbf{x}}} c(\mathbf{x}, \tilde{\mathbf{x}})$$

$$c(\mathbf{x}, \tilde{\mathbf{x}}) = \lambda_{xy} \|\mathbf{x}, \tilde{\mathbf{x}}\|_2 + \lambda_{\text{goal}} \|\mathbf{x}_T, \tilde{\mathbf{x}}_T\|_2 + \sum_{\phi \in \Phi} \phi(\mathbf{x})$$





模块化端到端系统

UniAD

OccFormer

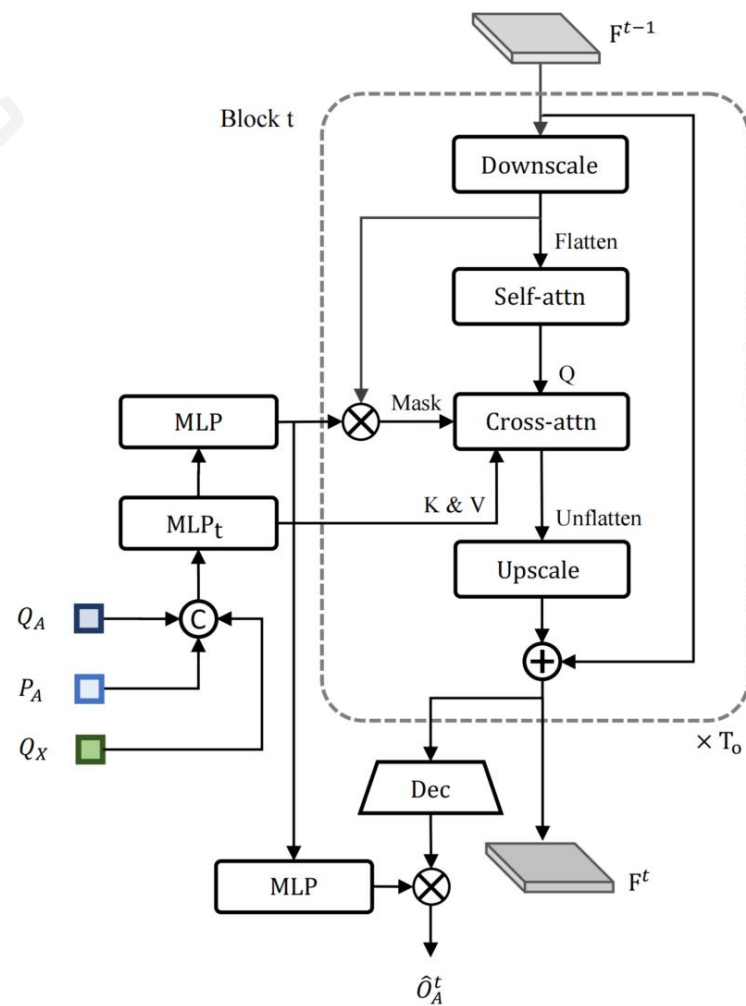
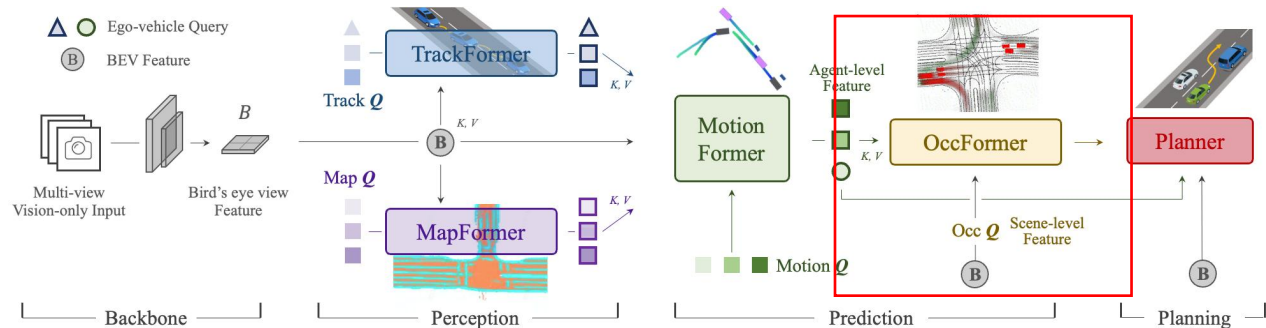
- 每个timestamp的Occ结果由连续的transformer block得到
- 融合track query、position embedding、motion query构建agent feature

$$G^t = \text{MLP}_t([Q_A, P_A, Q_X]), t = 1, \dots, T_o$$

- agent feature与BEV feature点乘，直接得到实例级Occ

$$\hat{O}_A^t = U^t \cdot F_{\text{dec}}^t$$

- 实例级Occ在计算过程中充当attention mask，以保持时序一致性
- 通过降采样提高计算效率





模块化端到端系统

UniAD

Planner

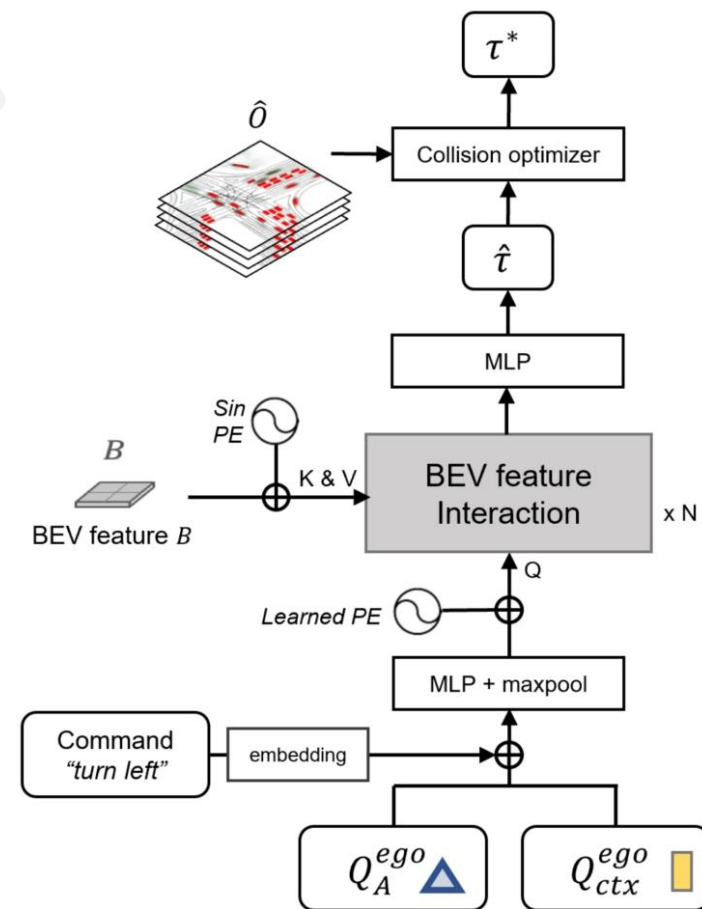
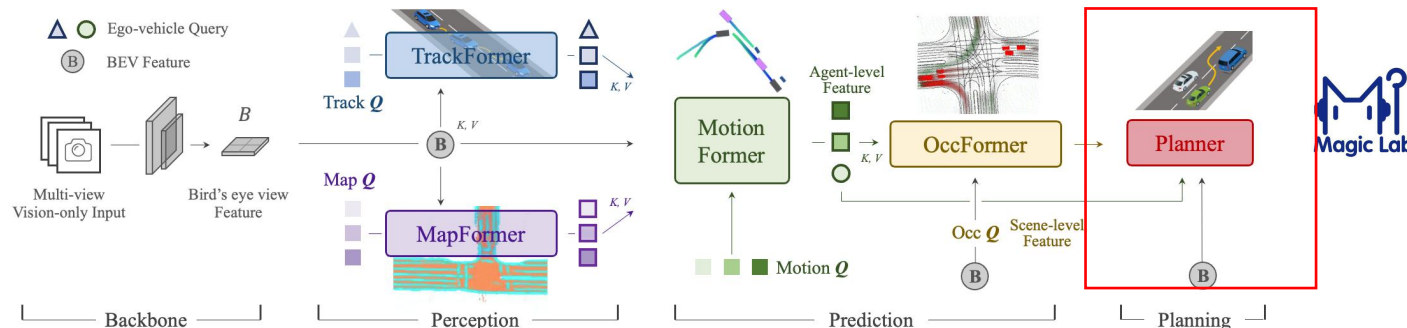
- 结合TrackFormer、MapFormer的ego query先验、command作为planning query
- 与BEV feature图进行attention, 预测轨迹
- 利用OccFormer预测的Occ结果, 进行避免碰撞的后处理

$$\tau^* = \arg \min_{\tau} f(\tau, \hat{\tau}, \hat{O})$$

$$f(\tau, \hat{\tau}, \hat{O}) = \lambda_{\text{coord}} \|\tau, \hat{\tau}\|_2 + \lambda_{\text{obs}} \sum \mathcal{D}(\tau_t, \hat{O}^t)$$

$$\mathcal{D}(\tau_t, \hat{O}^t) = \sum_{(x,y) \in \mathcal{S}} \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{\|\tau_t - (x,y)\|_2^2}{2\sigma^2}\right)$$

- $\lambda_{\text{coord}} \|\tau, \hat{\tau}\|_2$ 使优化后的轨迹与原轨迹尽量接近
- $\sum \mathcal{D}(\tau_t, \hat{O}^t)$ 使轨迹远离occupancy grid, 避免碰撞



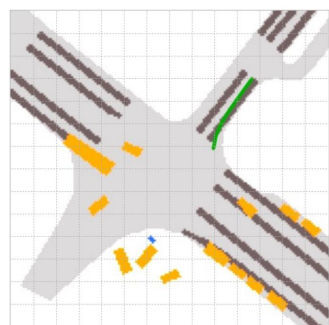


模块化端到端系统 架构优化

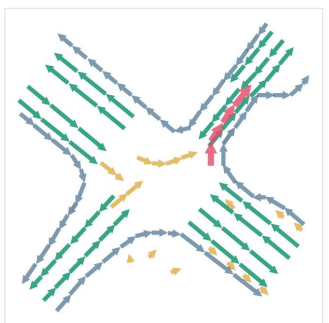


□ VAD

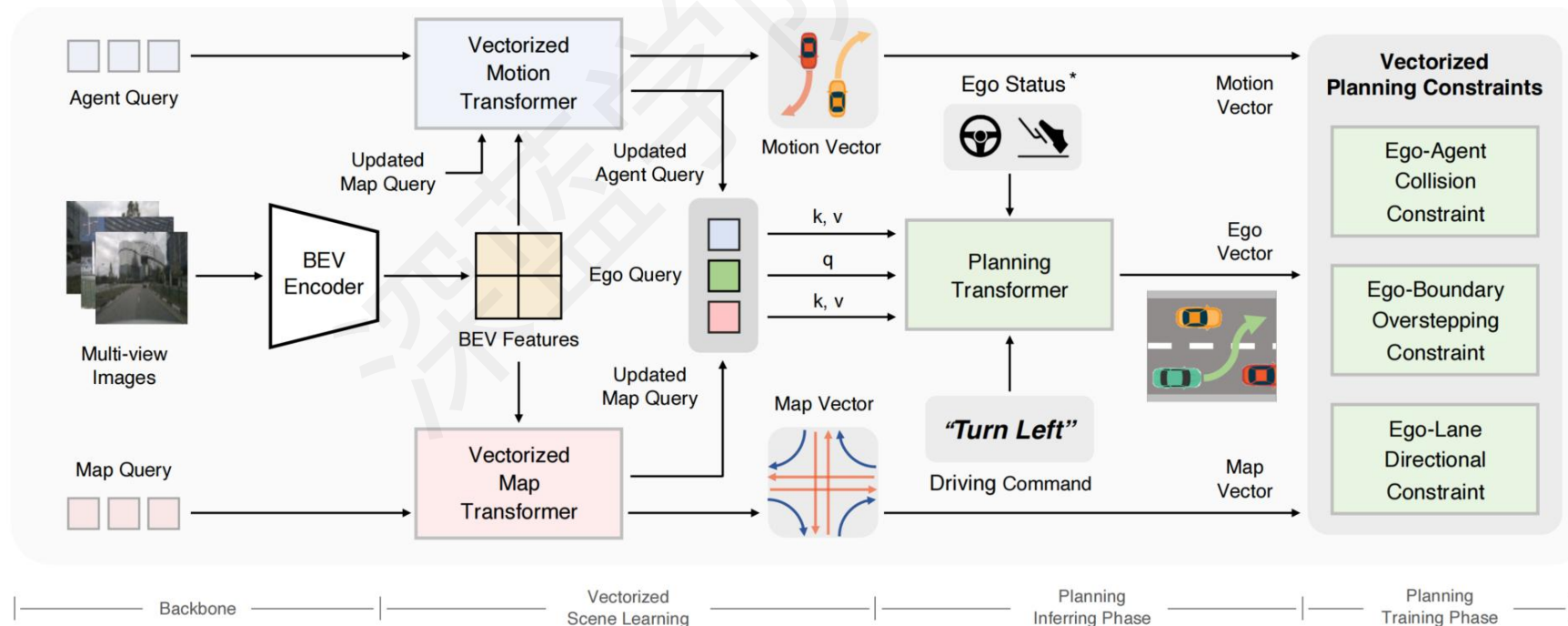
- 基于**矢量场景表征**的端到端自动驾驶，对整个驾驶场景进行矢量化建模，并利用矢量环境信息对自车规划轨迹进行约束
 - 矢量化建模：agent运动轨迹与地图元素使用矢量列表表示，摒弃计算密集型的密集栅格化和人工设计的后处理步骤
 - 优势：提高计算效率；增加实例级理解



(a) Rasterized Scene Representation



(b) Vectorized Scene Representation





模块化端到端系统 架构优化

□ 矢量化建模

- BEV Encoder用于编码输入的环视图像，并将其转化为鸟瞰图视角（BEV）下的特征图
- Vectorized Motion Transformer提取场景中的动态目标信息，实现动态目标检测和矢量化的轨迹预测

$$Q_a \longrightarrow \hat{V}_a \in \mathbb{R}^{N_a \times N_k \times T_f \times 2}$$

- Vectorized Map Transformer提取场景中矢量化的静态元素信息（如车道线，路沿和人行道）

$$Q_m \longrightarrow \hat{V}_m \in \mathbb{R}^{N_m \times N_p \times 2}$$

- Planning Transformer以隐式的动静态场景特征作为输入，提取其中与驾驶决策规划相关的信息，并完成自动驾驶车辆的轨迹规划

$$Q'_{ego} = \text{TransformerDecoder}(q, k, v, q_{pos}, k_{pos}),$$

$$q = Q_{ego}, k = v = Q_a,$$

$$q_{pos} = \text{PE}_1(p_{ego}), k_{pos} = \text{PE}_1(p_a).$$

$$Q''_{ego} = \text{TransformerDecoder}(q, k, v, q_{pos}, k_{pos}),$$

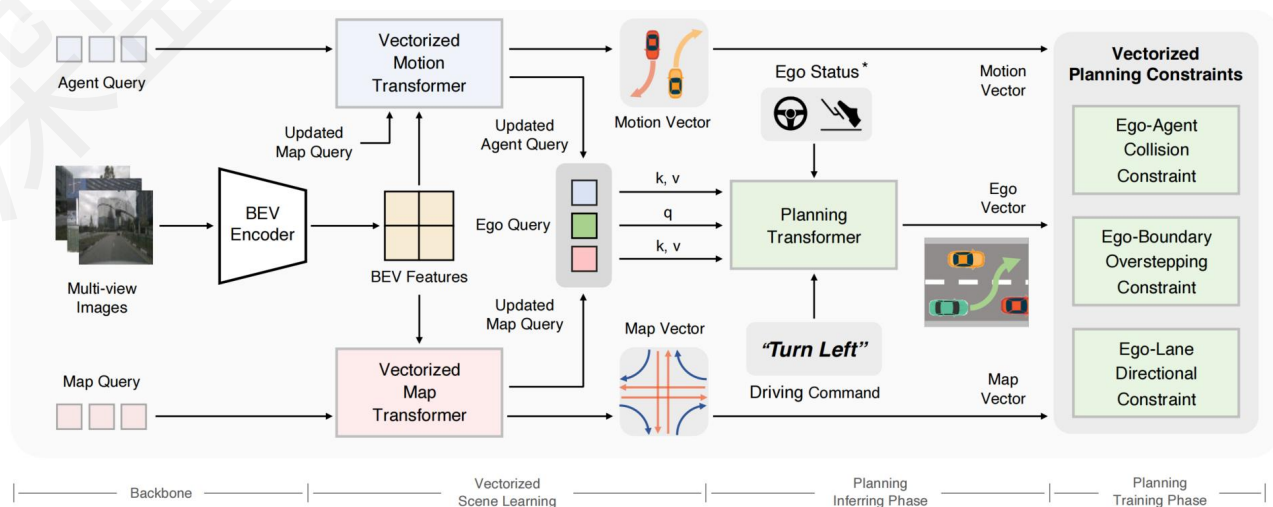
$$q = Q'_{ego}, k = v = Q_m,$$

$$q_{pos} = \text{PE}_2(p_{ego}), k_{pos} = \text{PE}_2(p_m).$$

$$\hat{V}_{ego} = \text{PlanHead}(ft = f_{ego}, cmd = c),$$

$$f_{ego} = [Q'_{ego}, Q''_{ego}, s_{ego}].$$

Method	L2 (m) ↓				Collision (%) ↓				Latency (ms)	FPS
	1s	2s	3s	Avg.	1s	2s	3s	Avg.		
NMP [†] [49]	-	-	2.31	-	-	-	1.92	-	-	-
SA-NMP [†] [49]	-	-	2.05	-	-	-	1.59	-	-	-
FF [†] [18]	0.55	1.20	2.54	1.43	0.06	0.17	0.07	0.43	-	-
EO [†] [24]	0.67	1.36	2.78	1.60	0.04	0.09	0.88	0.33	-	-
ST-P3 [19]	1.33	2.11	2.90	2.11	0.23	0.62	1.27	0.71	628.3	1.6
UniAD [21]	0.48	0.96	1.65	1.03	0.05	0.17	0.71	0.31	555.6	1.8
VAD-Tiny	0.46	0.76	1.12	0.78	0.21	0.35	0.58	0.38	59.5	16.8
VAD-Base	0.41	0.70	1.05	0.72	0.07	0.17	0.41	0.22	224.3	4.5
VAD-Tiny [‡]	0.20	0.38	0.65	0.41	0.10	0.12	0.27	0.16	59.5	16.8
VAD-Base [‡]	0.17	0.34	0.60	0.37	0.07	0.10	0.24	0.14	224.3	4.5





模块化端到端系统 架构优化



□ 规划约束

- 自车-他车碰撞约束：基于预定义的横向、纵向安全边界，约束自车轨迹于边界距离始终保留安全阈值以上

$$\mathcal{L}_{\text{col}} = \frac{1}{T_f} \sum_{t=1}^{T_f} \sum_i \mathcal{L}_{\text{col}}^{it}, i \in \{X, Y\},$$

$$\mathcal{L}_{\text{col}}^{it} = \begin{cases} \delta_i - d_a^{it}, & \text{if } d_a^{it} < \delta_i \\ 0, & \text{if } d_a^{it} \geq \delta_i. \end{cases}$$

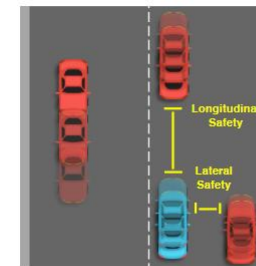
- 自车-边界越界约束：基于预测的矢量化边界线，约束自车规划轨迹始终在可行驶区域内

$$\mathcal{L}_{\text{bd}} = \frac{1}{T_f} \sum_{t=1}^{T_f} \mathcal{L}_{\text{bd}}^t,$$

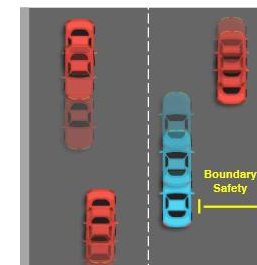
$$\mathcal{L}_{\text{bd}}^t = \begin{cases} \delta_{bd} - d_{bd}^t, & \text{if } d_{bd}^t < \delta_{bd} \\ 0, & \text{if } d_{bd}^t \geq \delta_{bd}, \end{cases}$$

- 自车-道路方向约束：基于预测的道路线方向，约束自车规划轨迹朝向与道路前进方向保持一致

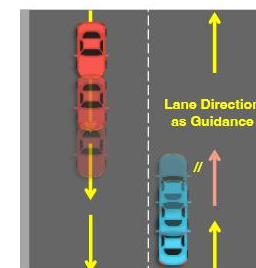
$$\mathcal{L}_{\text{dir}} = \frac{1}{T_f} \sum_{t=1}^{T_f} F_{\text{ang}}(\hat{v}_m^t, \hat{v}_{\text{ego}}^t)$$



Ego-Agent
Collision Constraint



Ego-Boundary
Overstepping Constraint



Ego-Lane
Directional Constraint





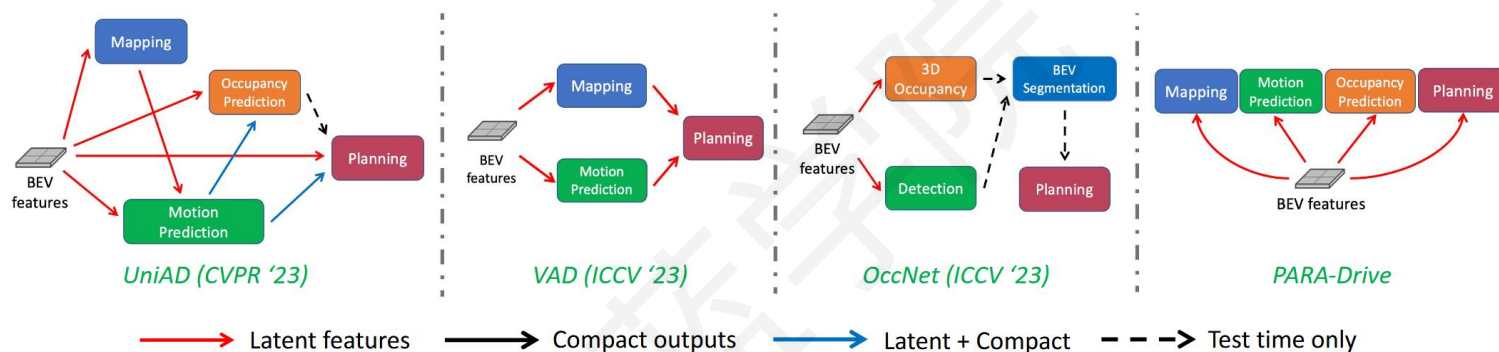
模块化端到端系统 架构优化



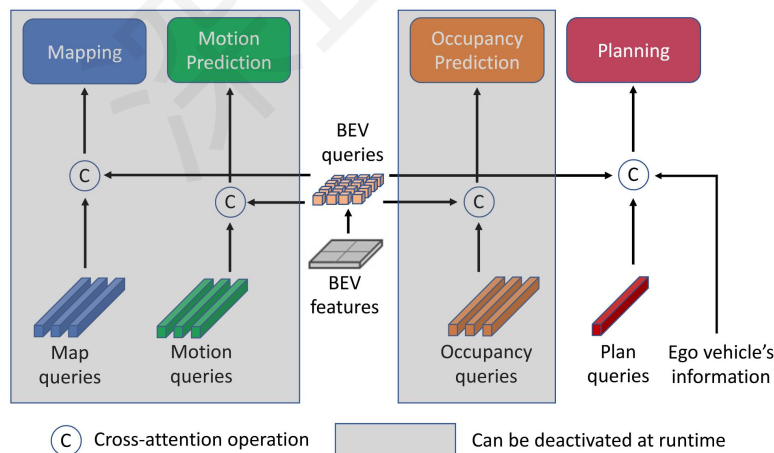
模块并行 PARA-Drive

- 构建了完全并行的端到端系统架构。多任务协同训练，性能SOTA，运行速度提升3倍。
- 架构简介，可解释性/运行速度调整灵活。

流程对比



算法架构



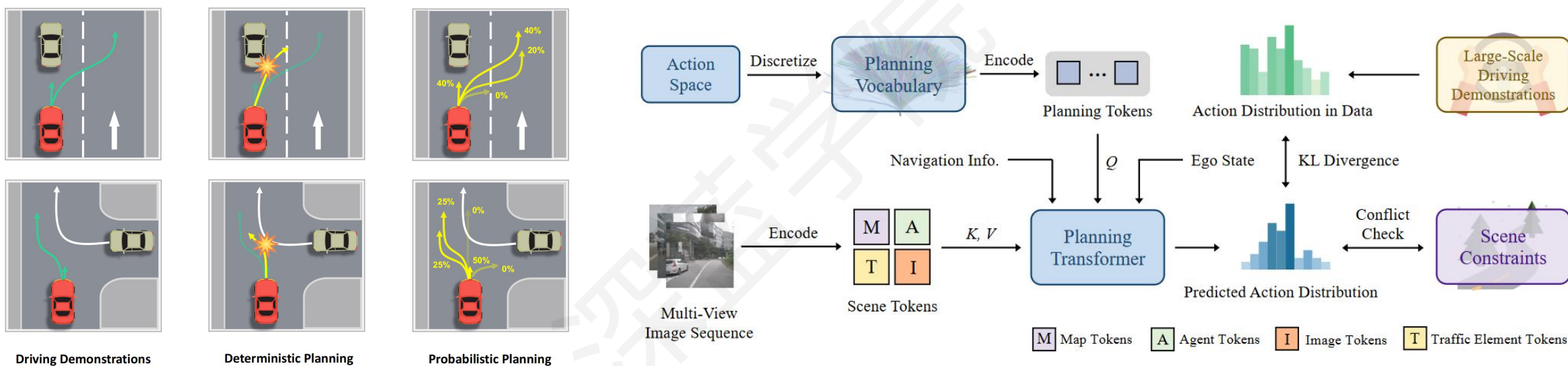


模块化端到端系统 架构优化



□ 离散轨迹概率规划 VAD v2

- 输出离散轨迹的概率分布，采样离散轨迹以控制车辆。
 - Planning Vocabulary: 收集驾驶数据集中的所有规划动作，并采用最远轨迹采样来选择 N 个具有代表性的动作



- 损失函数
 - 分布损失: 输出动作的概率分布和真实的分布之间的KL散度作为约束 $\mathcal{L}_{\text{distribution}} = D_{\text{KL}}(p_{\text{data}} || p_{\text{pred}})$.
 - 碰撞损失: 与其他物体/道路边界碰撞的轨迹视为负样本
 - 场景token损失: map、agent、traffic element模块分别对应任务监督





端到端系统评测

□ nuScenes开环评测

- L2 error

输出轨迹用3s内的6个轨迹点表示, 即 $\tau = [(x_1, y_1), \dots, (x_6, y_6)]$

对每个驾驶场景的输出轨迹 τ , L2 error被计算为:

$$l_2 = \left[\sqrt{(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2} \right]_{i=1}^6$$

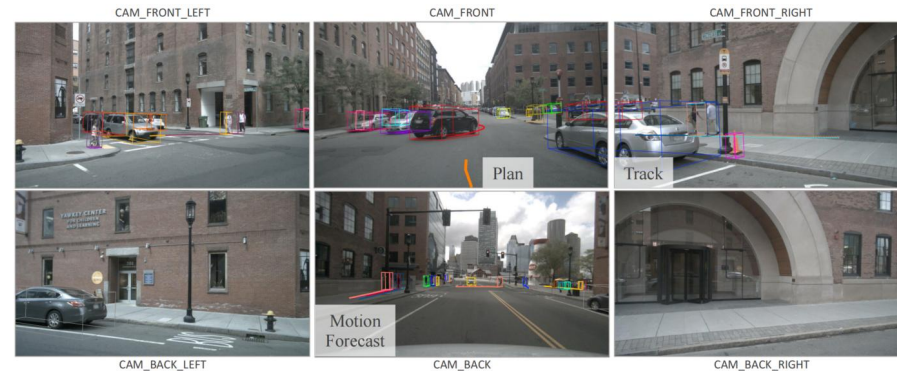
第 k 秒的L2 error为:

$$L_{2,k}^{uniad} = \bar{l}_2[2k], \quad L_{2,k}^{stp3} = \frac{\sum_{t=1}^{2k} \bar{l}_2[t]}{2k}$$

- 碰撞率 (collision rate)

$$C_k^{uniad} = C[2k], \quad C_k^{stp3} = \frac{\sum_{t=1}^{2k} C[t]}{2k}$$

其中, C 为每个时间步的总碰撞次数。





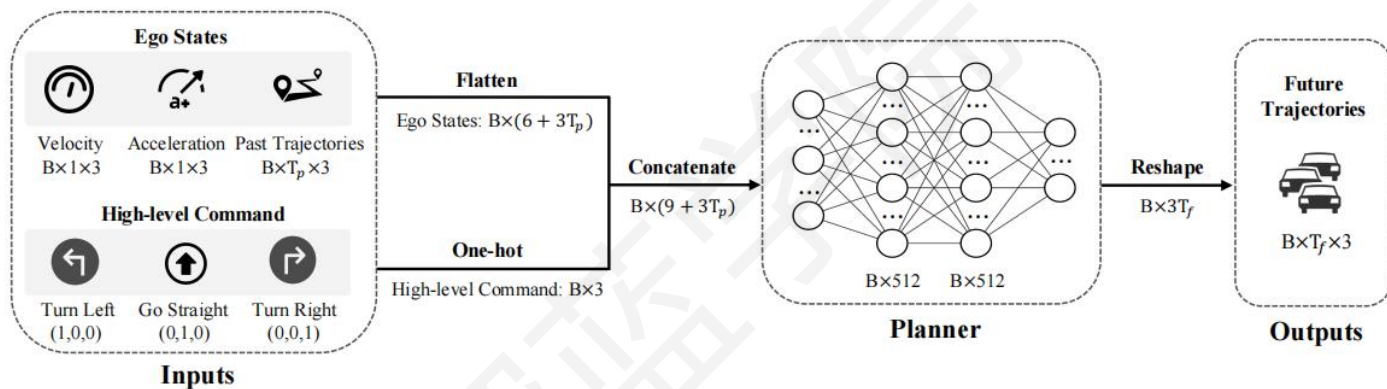
端到端系统评测



□ nuScenes开环规划性能受自车状态高度影响

- AD-MLP

仅依赖自车状态(ego status)，并且在现有的L2 error和碰撞率指标方面与最先进方法不相上下。



Method	Perception Information	High-level Command	Ego State			L2 (m) ↓				Collision (%) ↓			
			Velocity	Acceleration	Trajectory	1s	2s	3s	Avg.	1s	2s	3s	Avg.
NMP [18]	✓	✗	-	-	-	-	-	2.31	-	-	-	1.92	-
SA-NMP [18]	✓	✗	-	-	-	-	-	2.05	-	-	-	1.59	-
FF [6]	✓	✗	-	-	-	0.55	1.20	2.54	1.43	0.06	0.17	1.07	0.43
EO [10]	✓	✗	-	-	-	0.67	1.36	2.78	1.60	0.04	0.09	0.88	0.33
ST-P3 [7]	✓	✓	-	-	-	1.33	2.11	2.90	2.11	0.23	0.62	1.27	0.71
UniAD [8]	✓	✓	-	-	-	0.48	0.96	1.65	1.03	0.05	0.17	0.71	0.31
VAD-Tiny [9]	✓	✓	✓	✓	✓	0.20	0.38	0.65	0.41	0.10	0.12	0.27	0.16
VAD-Base [9]	✓	✓	✓	✓	✓	0.17	0.34	0.60	0.37	0.07	0.10	0.24	0.14
Ours	✗	✗	-	-	✓	0.53	0.91	1.48	0.97	0.17	0.46	0.83	0.49
	✗	✗	-	✓	✓	0.33	0.48	0.66	0.49	0.21	0.29	0.40	0.30
	✗	✗	✓	✓	✓	0.24	0.32	0.49	0.35	0.18	0.22	0.28	0.23
	✗	✓	✓	✓	✓	0.20	0.26	0.41	0.29	0.17	0.18	0.24	0.19



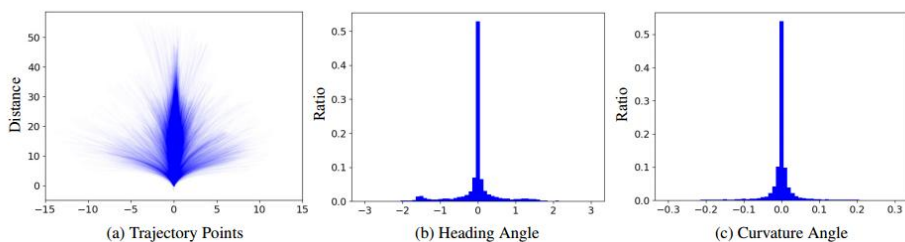


端到端系统评测

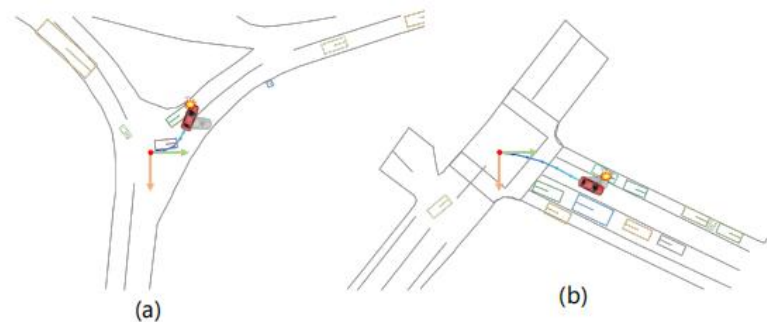


□ nuScenes开环评测存在的问题

- 数据集不平衡，在nuScenes训练集中自车在短时间范围内行驶时倾向于沿直线和小角度前进。



- 碰撞率结果受占用网格大小影响，且碰撞率计算时航向角保持不变。
- 现有评估指标不全面：一些高风险轨迹，如驶出道路可能在现有指标中不会受到严重惩罚。
- 自车状态偏见与驾驶逻辑相矛盾：在某些情况下，从现有的端到端自动驾驶框架中完全移除视觉输入，并不会显著降低规划行为的质量。这与基本的驾驶逻辑相矛盾。



Zhai J T, Feng Z, Du J, et al. Rethinking the open-loop evaluation of end-to-end autonomous driving in nusenes[J]. arXiv preprint arXiv:2305.10430, 2023.

Li Z, Yu Z, Lan S, et al. Is ego status all you need for open-loop end-to-end autonomous driving?[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024: 14864-14873.





□ 评测指标优化

- 路缘碰撞率 Curb Collision Rate
 - 表示离开可驾驶区域的可能性。nuScenes上某些注释的道路边界确实是可穿越的，并且在特定条件下，真值轨迹可能与这些边界相交。
 - 每个时刻的碰撞并非相互独立

$$CR(t) = \sum_{i=0}^N \Pi_i / N, N = t/0.5 \longrightarrow CR(t) = (\sum_{i=0}^N \Pi_i) > 0, N = t/0.5$$

其中， N 是 t 秒的时间步长数， Π_i 代表在第 i 步自车是否与其他智能体相交

- 轨迹平滑度 Trajectory Smoothness
 - 将模型预测的不同的轨迹点看作一个分布，计算轨迹点的方差作为平滑度量。

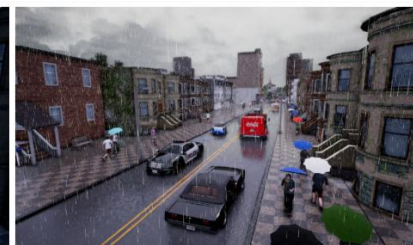
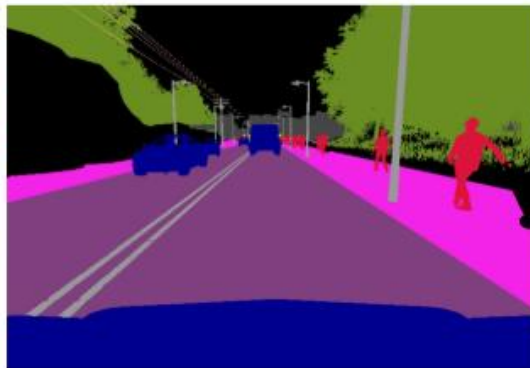
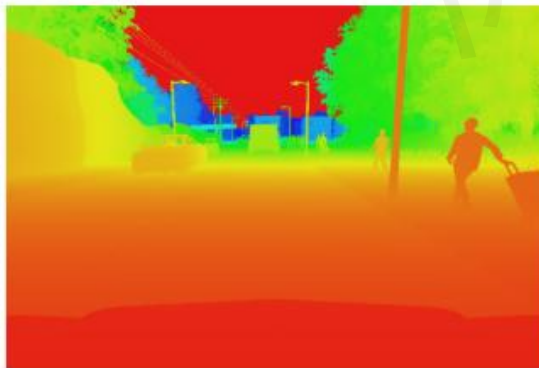




端到端系统评测

▣ CARLA仿真器闭环评测

- CARLA仿真器
 - 通过简单的接口与动态世界交互
 - 可布置的多种静态与动态对象
 - 通过控制器实现非玩家车辆的行为逼真度
 - 随机生成具有视觉多样性的行人
 - 不同大气条件和照明模式
 - 种类丰富的传感器
 - 提供与智能体状态和交通规则合规性测量
 - 支持自动驾驶系统开发、训练和详细性能分析
 - Route Completion Ratio; Infraction Score; Driving Score.





端到端系统评测



ReasonNet

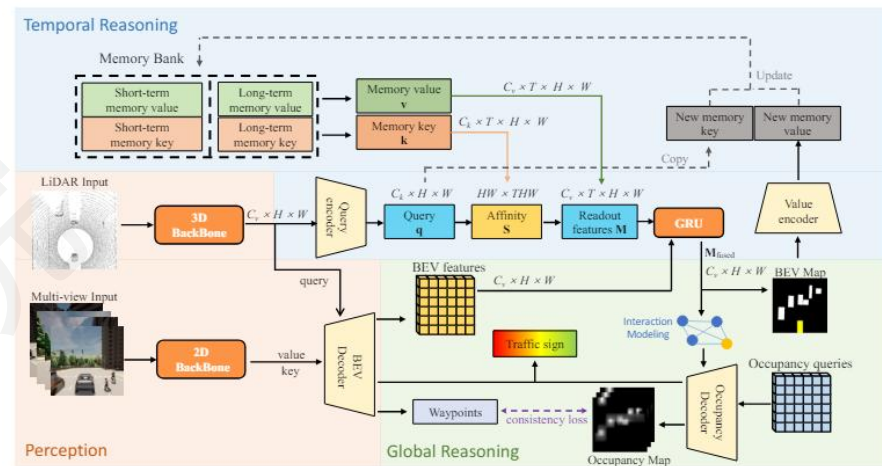
- 利用驾驶场景的时间和全局信息，推理物体的时间行为，提高整体感知性能

- 时间推理模块：

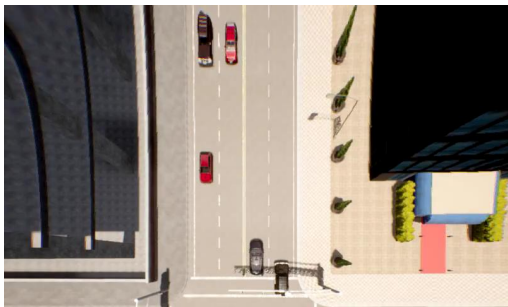
有效融合不同帧的信息，对驾驶场景做出全面理解，对驾驶场景的未来演变做出高保真预测，从而提高驾驶性能

- 基于transformer的全局推理模块：

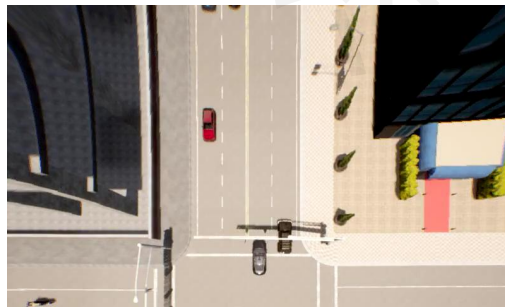
充分融合环境和物体的信息，并分析它们的交互，处理长尾分布中的罕见不利事件，以更好地理解场景



包含4个遮挡场景的遮挡驾驶仿真基准（Driving in Occlusion Simulation benchmark，简称DOS），每个场景有25个案例，作为端到端自动驾驶领域中全面的遮挡事件评估基准。



Parked cars



Sudden brake



Left turn



Red light infringement



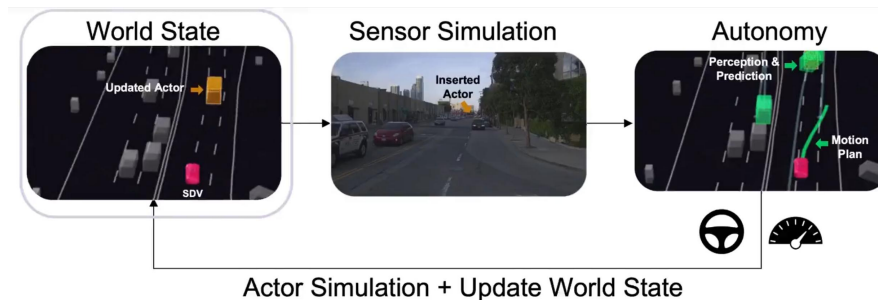
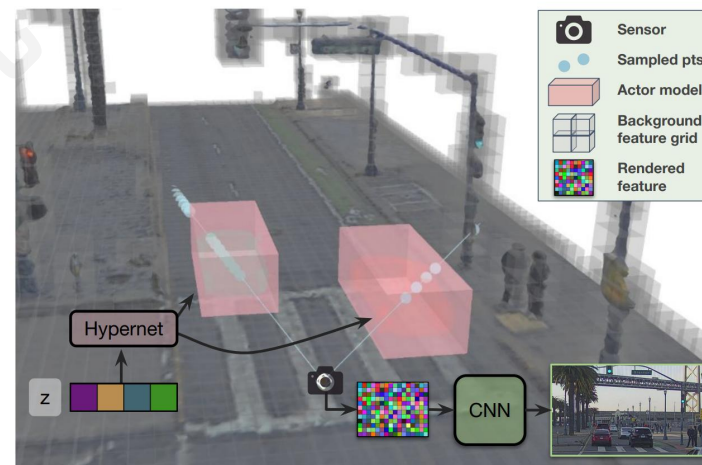
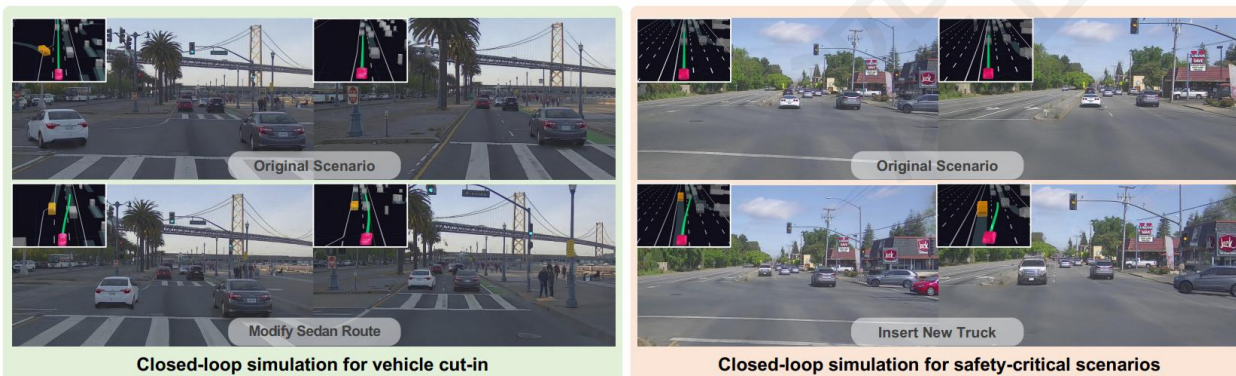
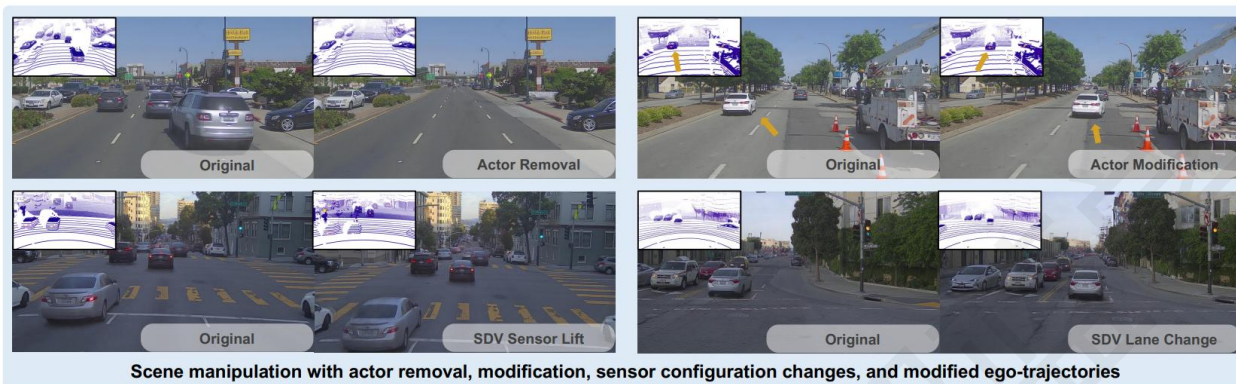


端到端系统评测



UniSim仿真器闭环评测

- Carla仿真器依赖数字资产构建，渲染真实性、多样性不足，存在较大Sim2Real Gap；
- UniSim使用采集的真实数据，基于NeRF重建驾驶场景，生成罕见危险场景，并支持闭环仿真。



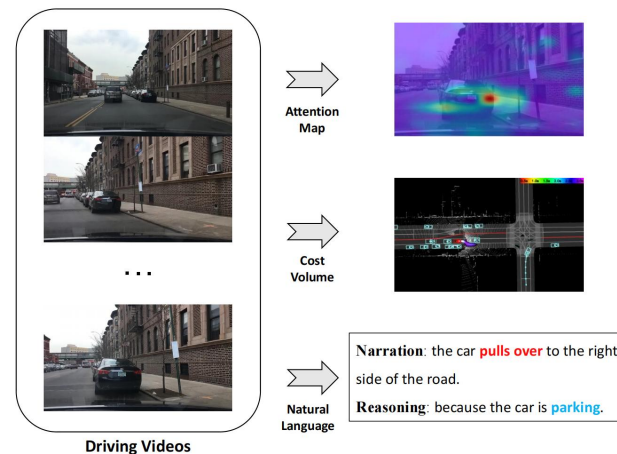
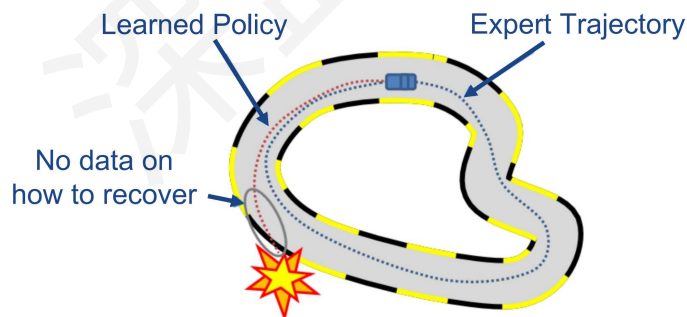


端到端自动驾驶



□ 局限与挑战

- 多模态输入如何有效融合，尤其是传感器与语言模态之间；
- 高精地图在端到端系统中如何有效建模表征；
- 规划部分对全量感知信息的有效提取需要进一步设计；
- 如何更好地处理长尾物体感知与长尾场景决策；
- 端到端算法在测试场景上闭环规划的OOD (Out of Distribution)问题如何解决；
- 历史规划信息和当前感知信息在当前规划时需要更好权衡；
- 端到端系统在不同交通情景下的行为可解释性需要增强。



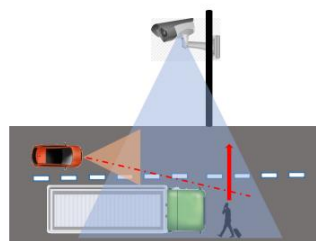
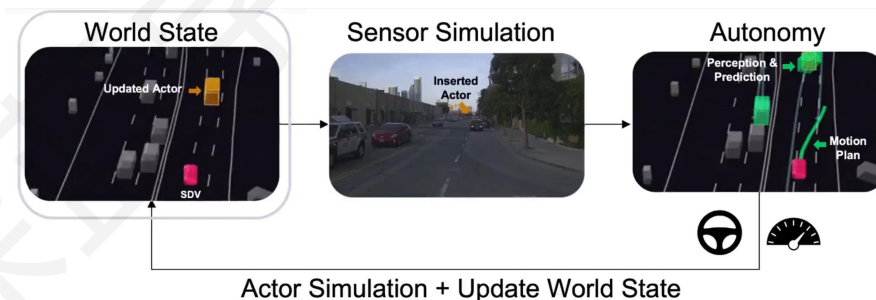
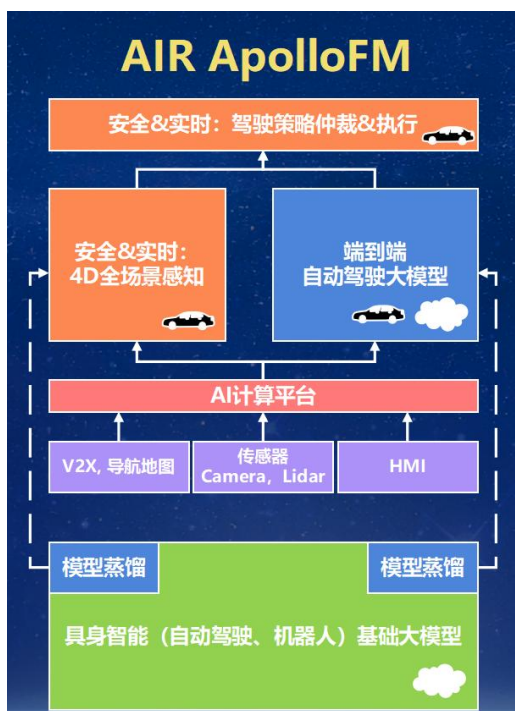
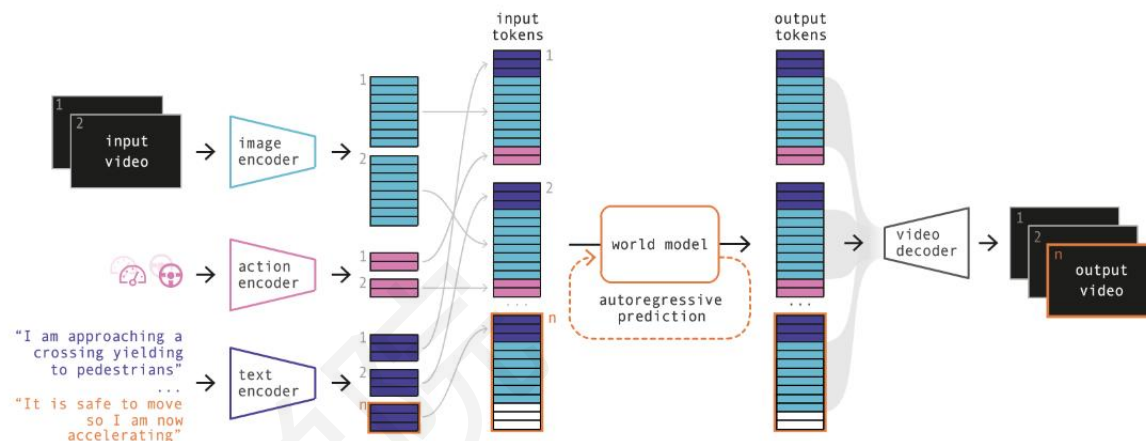


端到端自动驾驶

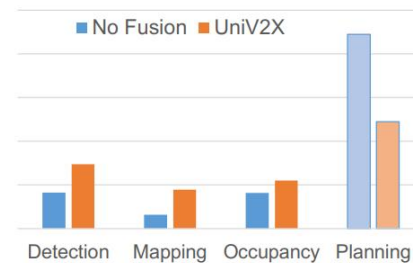


发展趋势

- 多模态大模型+自动驾驶
- World Model
- 端到端闭环仿真
- V2X+端到端



(a) VICAD



(b) Evaluation Results.





大纲



端到端自动驾驶



自动驾驶与多模态大模型

深蓝学院



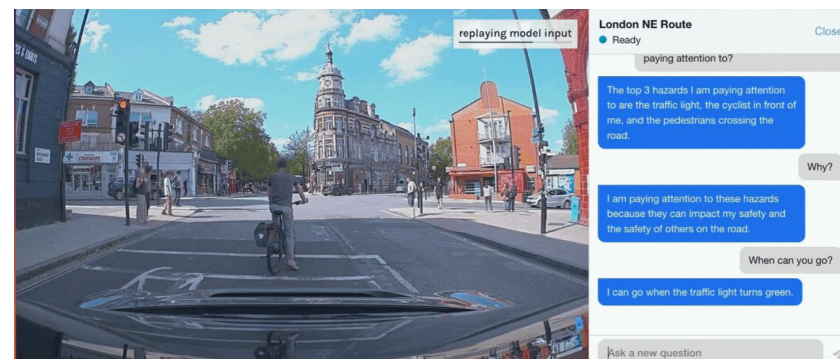
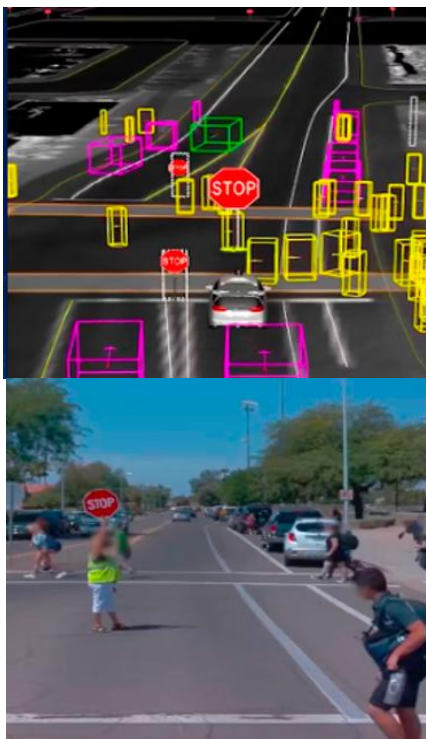


自动驾驶与多模态大模型



多模态大模型之于自动驾驶的优势

- 常识与推理能力——应对长尾场景，解决OOD问题
- 开集场景理解——提取全量感知信息，实现长尾物体感知
- 文字理解能力——提高可解释性和人机交互能力



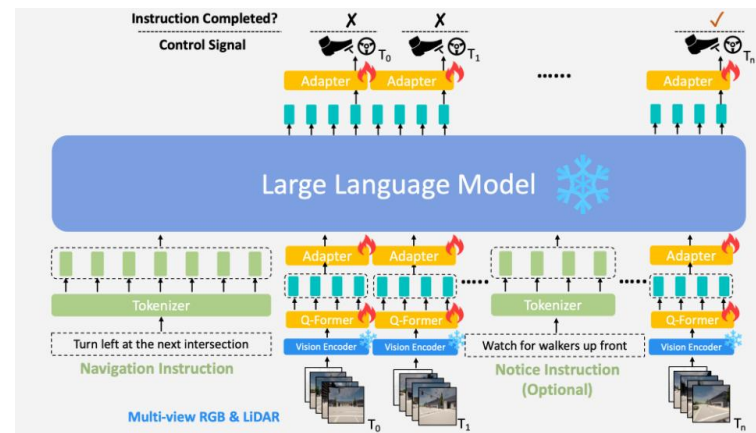
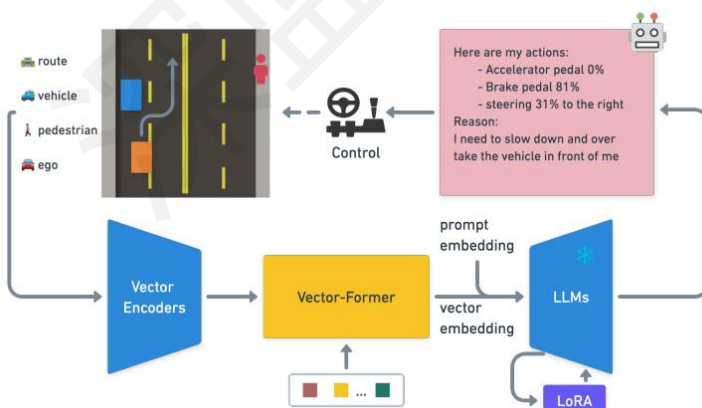
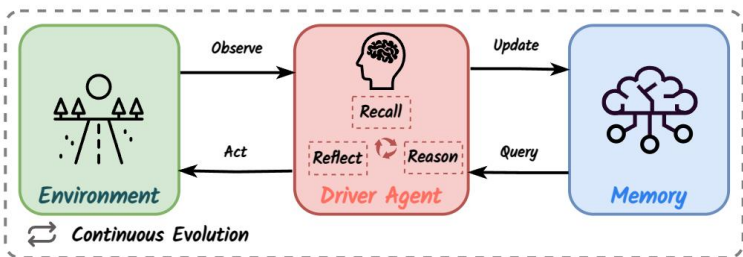
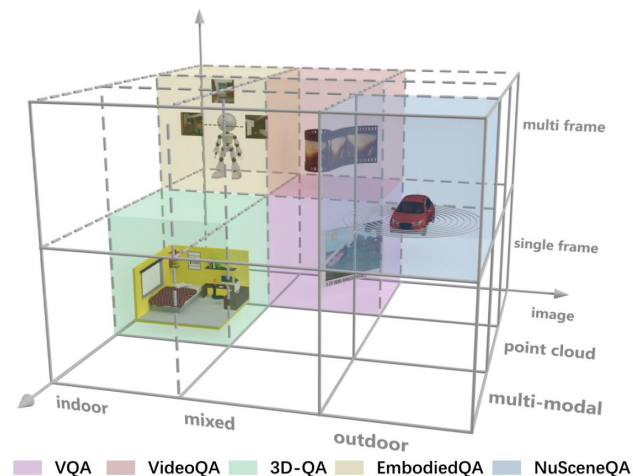
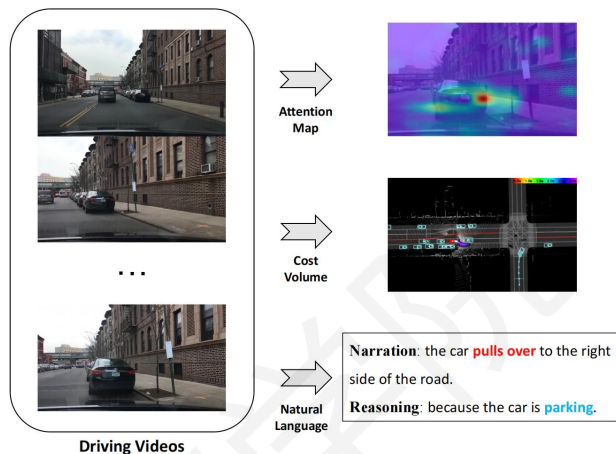


自动驾驶与多模态大模型



发展脉络与趋势

- 自动驾驶语言可解释性
- 自动驾驶文本数据集
- 基于场景文本特征的规划模型
- 场景特定特征对齐的规划模型
- 闭环多模态大模型
- 大模型特性探索



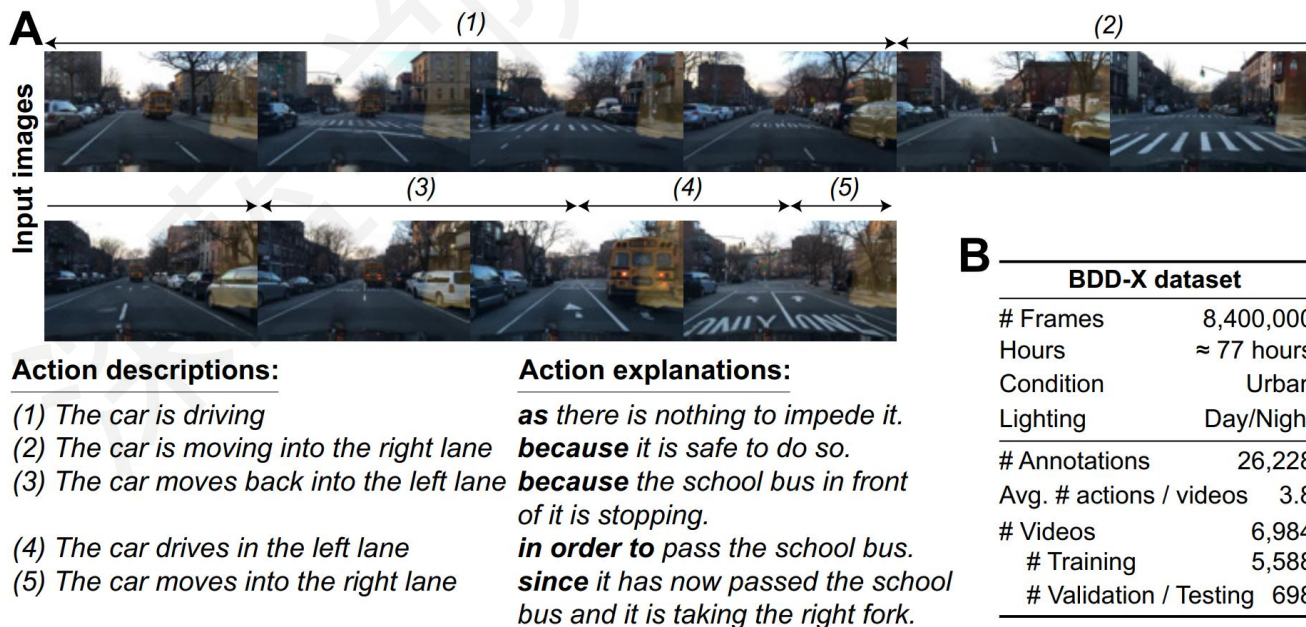


自动驾驶语言可解释性



□ BDD-X Dataset

- Berkeley Deep Drive-X 自动驾驶行为描述与解释数据集
- 可解释自动驾驶
 - 端到端驾驶数据: video; control
 - 行为描述与解释
 - 用户接受度
 - 车辆行为理解和推断
 - 用户交互



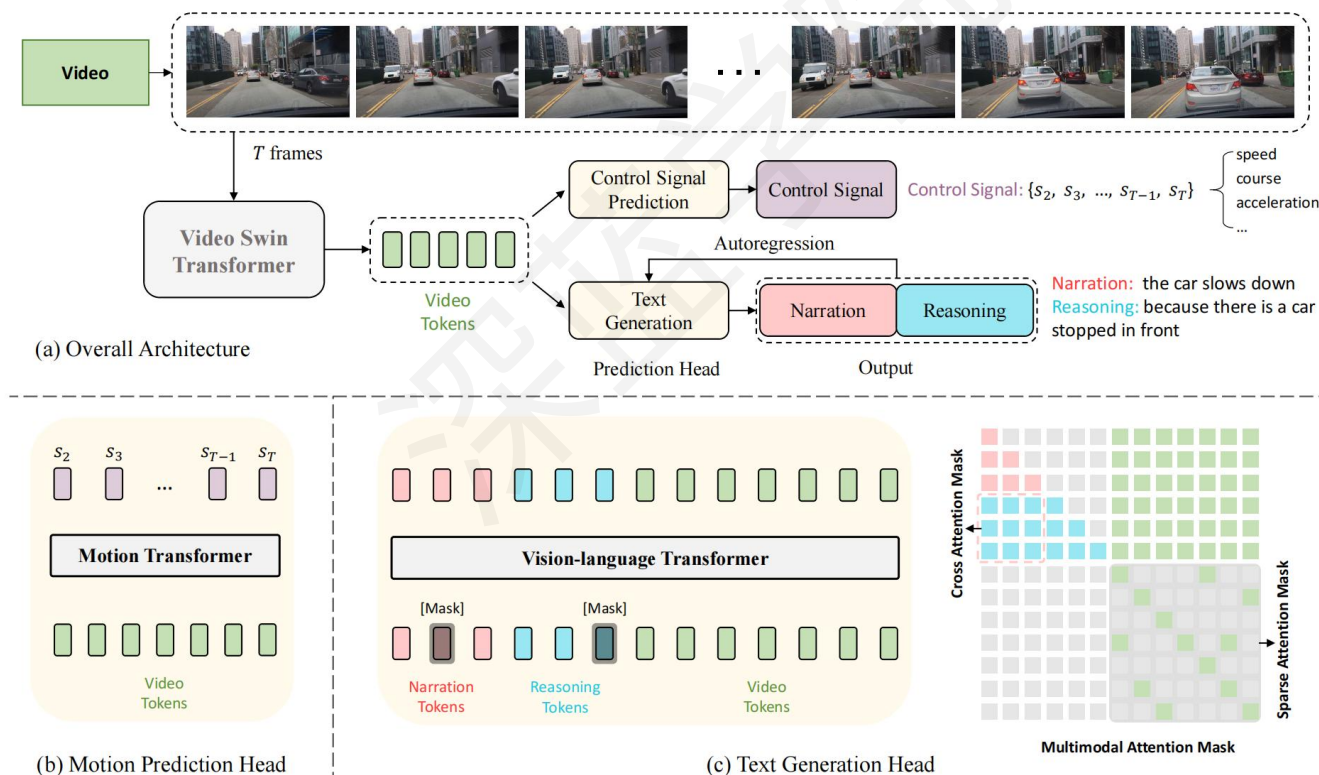


自动驾驶语言可解释性



ADAPT

- Driving Caption Generation与Control Signal Prediction多任务协同训练
- 借助Swinbert的多模态模型能力，实现有效的驾驶行为描述与解释，提高自动驾驶可解释性
- Reasoning与Narration之间Causal Attention，增强reasoning合理性



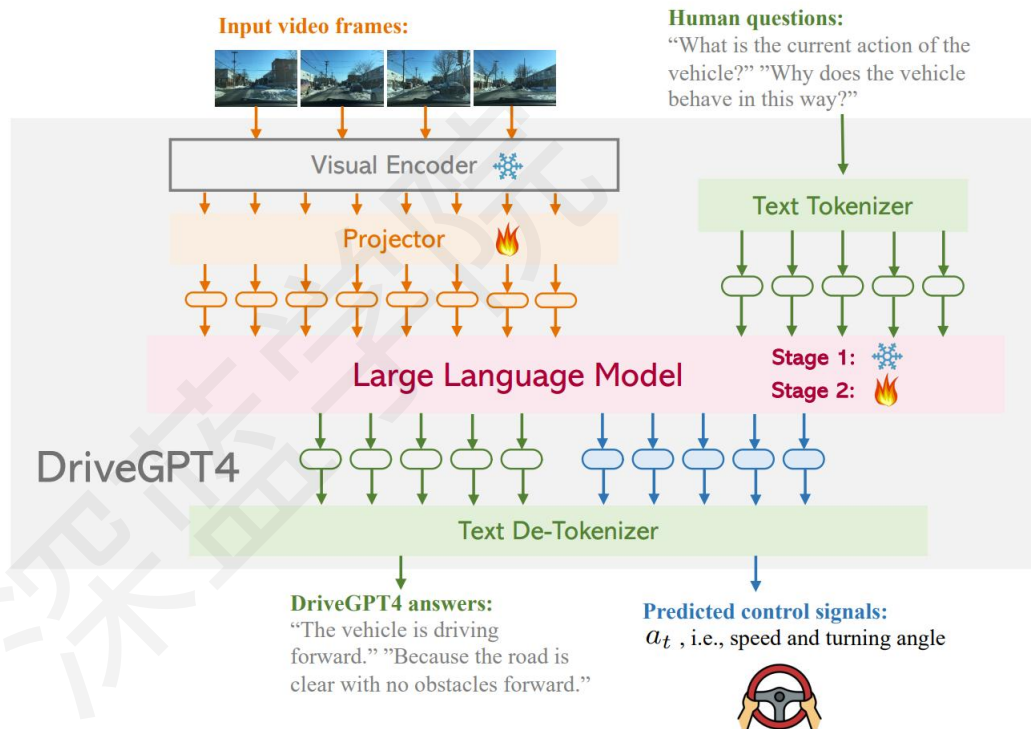


自动驾驶语言可解释性



□ DriveGPT4

- DriveGPT4通过微调LLMs，可以同时生成对人类询问的响应并预测控制信号。
- 凭借大模型的文本生成能力，自动驾驶文本解释不局限于BDD-X数据集定义的形式
- 仅对固定的QA进行训练将降低LLMs的推理能力，并使其无法回答其他格式的问题。因此，DriveGPT4使用了40K ChatGPT生成的额外对话进行训练。



Stage 1: Pretraining (Alignment)

595K Videos

703K Images

Stage 2: Mix-finetune (Instruction-tuning)

73K Videos

150K Images

+

16K
BDD-X QAs

40K ChatGPT QAs

ChatGPT



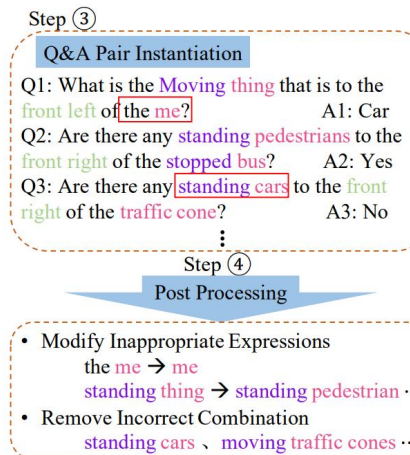
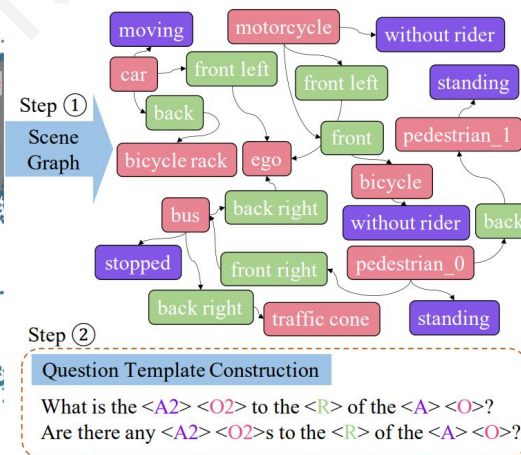
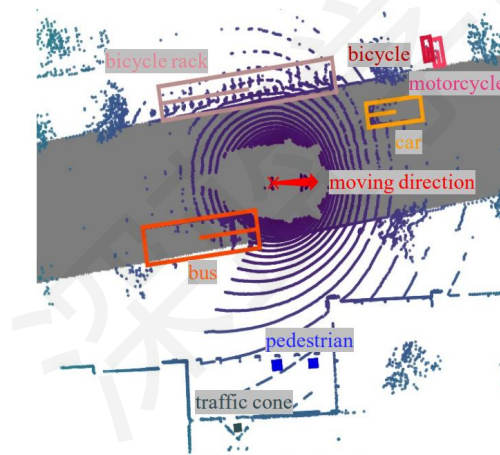
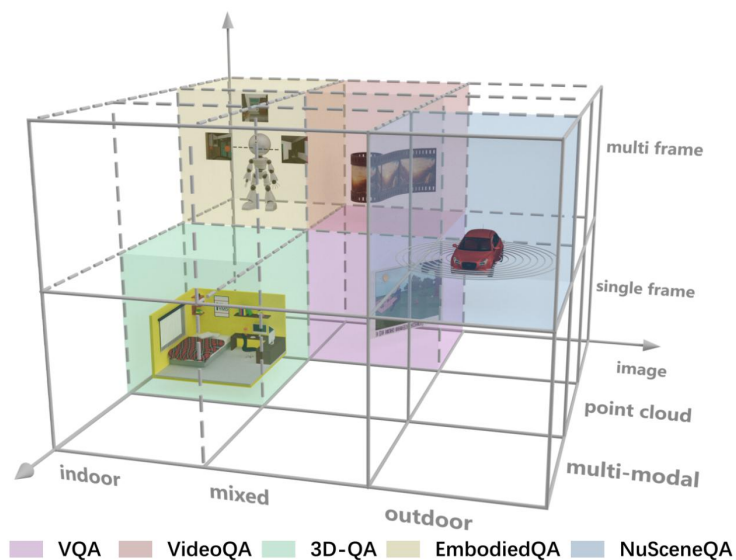


自动驾驶文本数据集



□ NuScenes-QA

- 自动驾驶场景下多模态、时序的视觉问答任务/数据集





自动驾驶文本数据集

□ NuScenes-QA

- 自动驾驶场景下多模态、时序的视觉问答任务/数据集

- 场景图构建

- 6种位置关系: front, back, front left, front right, back left, back right

$$\theta = \cos^{-1} \frac{(B_1[:2] - B_2[:2]) \cdot V_{ego}[:2]}{\|B_1[:2] - B_2[:2]\| \|V_{ego}[:2]\|}$$
$$relation = \begin{cases} front & \text{if } -30^\circ < \theta \leq 30^\circ \\ front\ left & \text{if } 30^\circ < \theta \leq 90^\circ \\ front\ right & \text{if } -90^\circ < \theta \leq -30^\circ \\ back\ left & \text{if } 90^\circ < \theta \leq 150^\circ \\ back\ right & \text{if } -150^\circ < \theta \leq -90^\circ \\ back & \text{else.} \end{cases}$$

- 问题模板设计

- 5类/66个问题模板: existence, counting, query-object, query-status, comparison

- <A> <O> <R>: attribute, object, relation

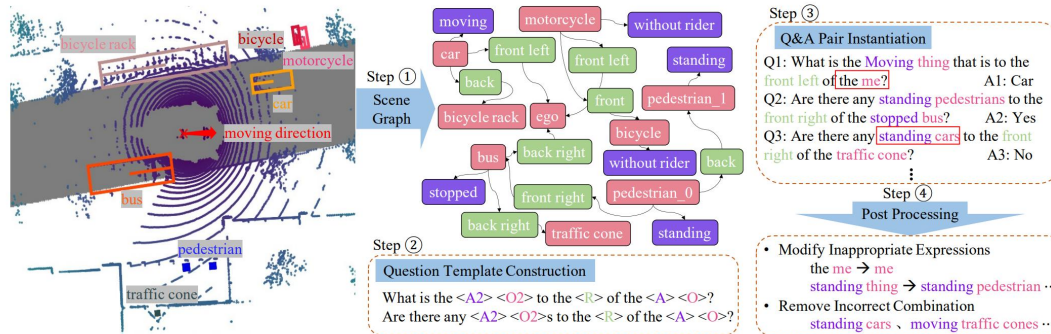
- zero-hop / one-hop

- spatial reasoning

Type	Number of Hop	Template
Existence	0	Are there any <A><O>s?
		Are any <A><O>s visible?
		Are there any other <A2><O2>s that in the same status as the <A><O>?
		Is there another <A2><O2> that has the same status as the <A><O>?
		Are there any other <A2><O2>s of the same status as the <A><O>?
	1	Is there another <A2><O2> of the same status as the <A><O>?
		Are there any <A2><O2>s to the <R> of the <A><O>?
		There is a <A><O>; are there any <A2><O2>s to the <R> of it?
		Are there any other <A3><O3>s that in the same status as the <A2><O2> [that is] to the <R> of the <A><O>?
		Is there another <A3><O3> that has the same status as the <A2><O2> [that is] to the <R> of the <A><O>?
		Are there any other <A3><O3>s of the same status as the <A2><O2> [that is] to the <R> of the <A><O>?
		Is there another <A3><O3> of the same status as the <A2><O2> [that is] to the <R> of the <A><O>?

- 问答对生成和过滤

- 去除不合理/退化的问答对, 如<A>+<O> 对应 “standing cars”/“parked pedestrians”



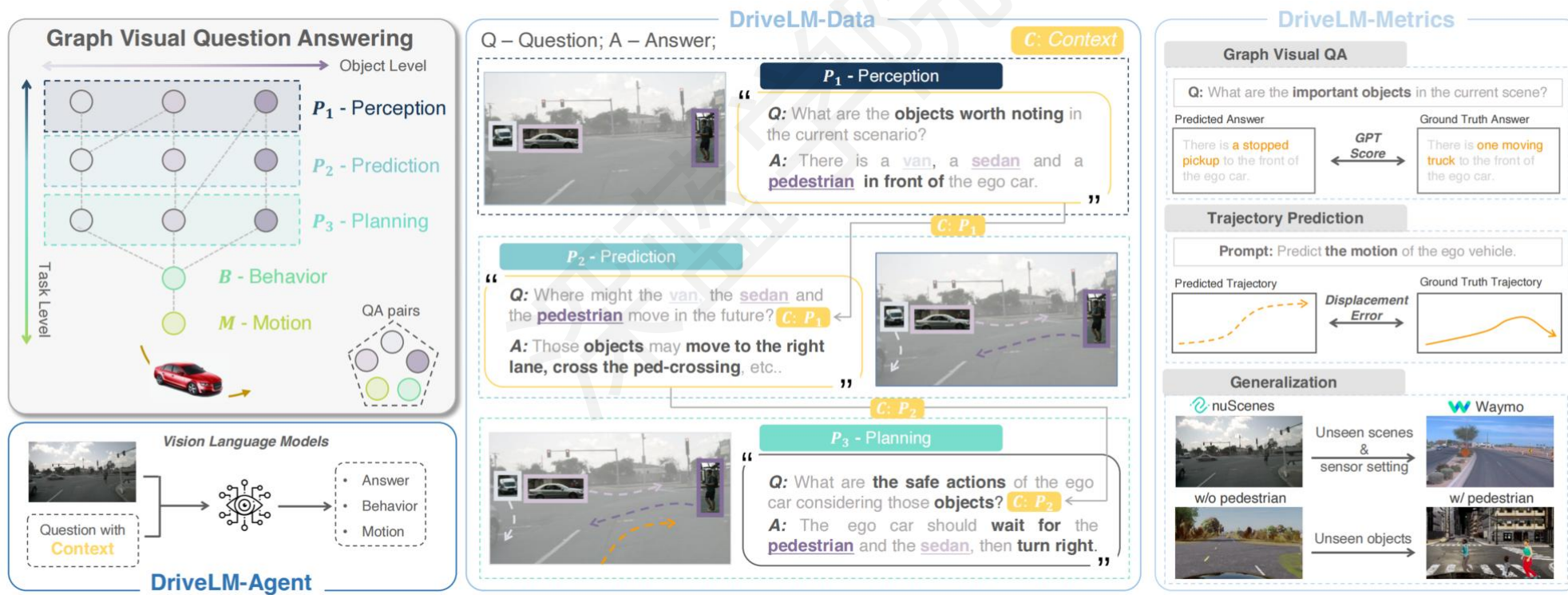


自动驾驶文本数据集



DriveLM

- 单轮VQA -> Graph VQA
- 问答对针对自动驾驶不同阶段进行标注：感知、预测、规划、行为决策、运动状态
- 图结构建模QA对上下文逻辑关系



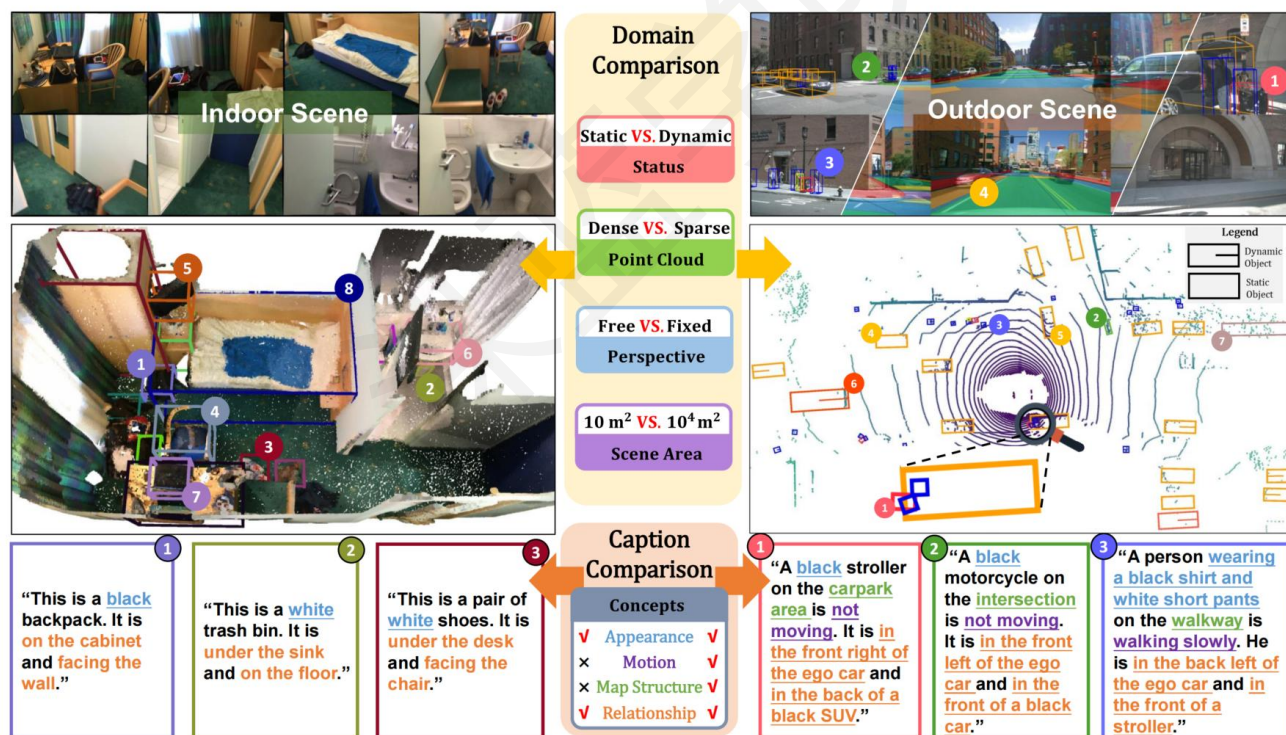


自动驾驶文本数据集



□ TOD³Cap

- 面向室外场景的3D密集描述
 - 从Dense Capturing任务（检测+描述）出发
 - 解决室外场景（1）动态;（2）稀疏点云;（3）固定有限视角;（4）更大关注范围 等问题
 - 标注：类别和视觉属性、环境定位、运动状态、位置关系



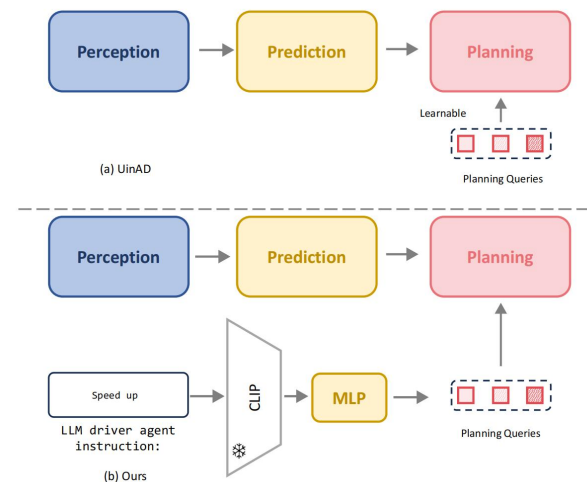
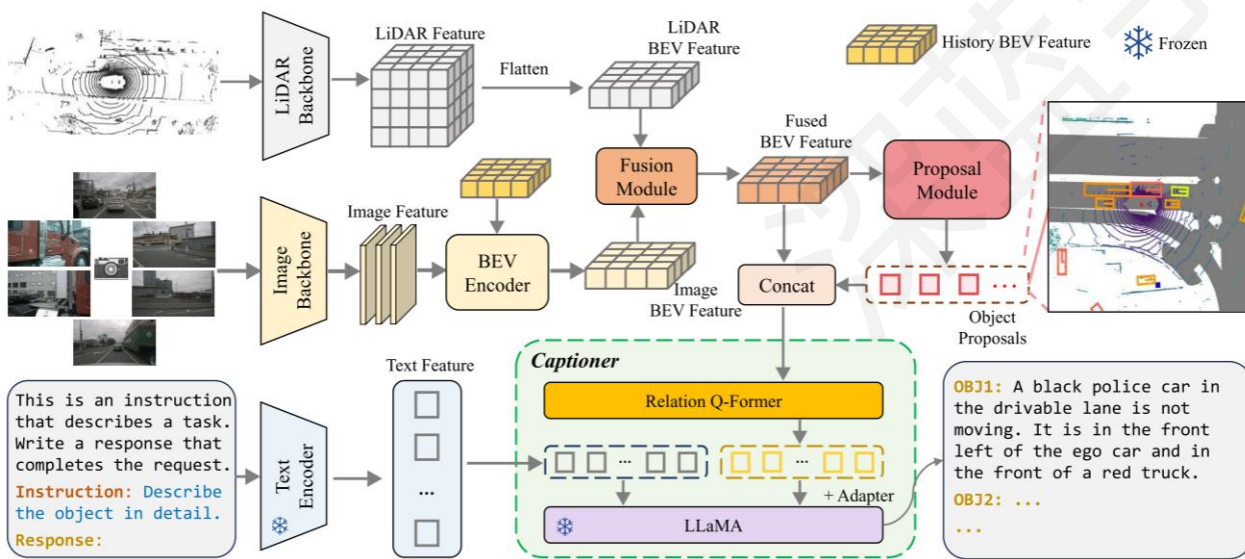
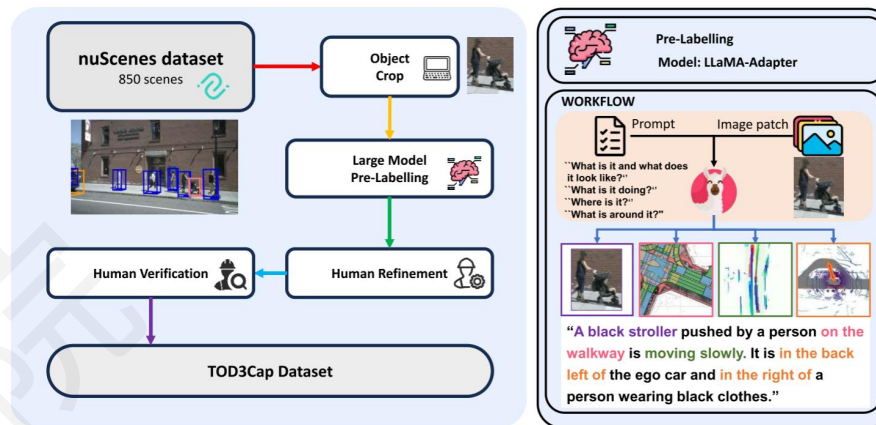


自动驾驶文本数据集



□ TOD³Cap

- 面向室外场景的3D密集描述
 - 半自动标注：物体框选，投影裁剪，MLLM自动标注，人工校验
 - 描述：检测器query与LLM实现特征对齐，描述结果准确可泛化
 - 端到端：caption结果/中间表达易嵌入端到端网络模块，提高性能





基于场景文本特征的规划模型

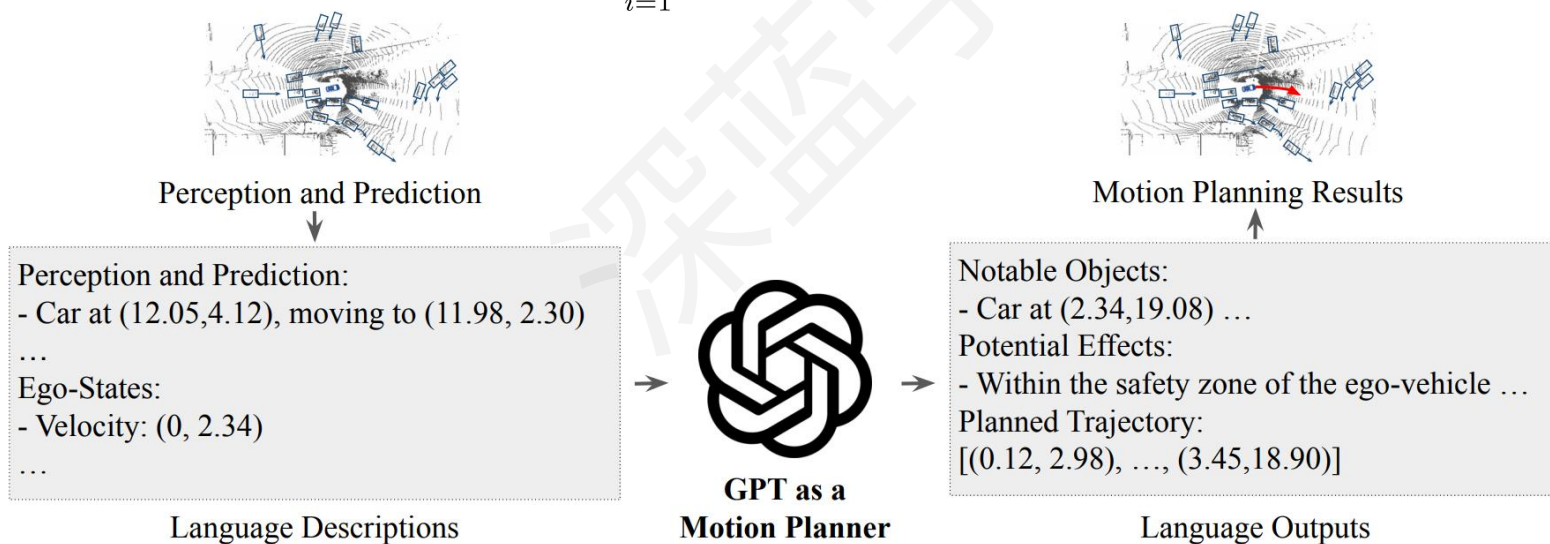


□ GPT-Driver

- 使用GPT直接以文本形式进行规划
- 将观察和自我状态转换为语言提示，引导LLM在自然语言中产生规划轨迹及其决策过程。

$$\mathcal{T} = K(\{(x_1, y_1), \dots, (x_t, y_t)\}) = \{w_1, \dots, w_n\}$$

$$\mathcal{L}_{LM} = - \sum_{i=1}^N \log P(\hat{w}_i | w_1, \dots, w_{i-1})$$





基于场景文本特征的规划模型



□ GPT-Driver

- 输入

参数化形式输入感知和观察结果、自车状态、历史轨迹等

- 输出

思维链推理过程

文本形式的预测轨迹

****Autonomous Driving Planner****

Role: You are the brain of an autonomous vehicle. Plan a safe 3-second driving trajectory. Avoid collisions with other objects.

Context

- Coordinates: X-axis is perpendicular, and Y-axis is parallel to the direction you're facing. You're at point (0,0).
- Objective: Create a 3-second route using 6 waypoints, one every 0.5 seconds.

Inputs

1. Perception & Prediction: Info about surrounding objects and their predicted movements.
2. Ego-States: Your current state including velocity, heading angular velocity, can bus data, heading speed, and steering signal.
3. Historical Trajectory: Your past 2-second route, given by 4 waypoints.
4. Mission Goal: High-level goal for the next 3 seconds.

Task

- Thought Process: Note down critical objects and potential effects from your perceptions and predictions.
- Action Plan: Detail your meta-actions based on your analysis.
- Trajectory Planning: Develop a safe and feasible 3-second route using 6 new waypoints.

Output

- Thoughts:
 - Notable Objects
 - Potential Effects
- Meta Action
- Trajectory (MOST IMPORTANT):
 - [(x1,y1), (x2,y2), ..., (x6,y6)]

Perception and Prediction:

- **animal** at (-1.93,7.00), moving to (-2.31,10.89).
- **car** at (-8.67,0.12), moving to (-8.50,-0.08).
- **adult** at (-1.21,6.78), moving to (-1.29,10.48).

Ego-States:

- Velocity (vx,vy): (0.00,1.46)
- Heading Angular Velocity (v_yaw): (-0.00)
- Acceleration (ax,ay): (0.01,-0.15)

Historical Trajectory (last 2 seconds): [(-0.00,-6.74), (-0.03,-4.73), (-0.03,-3.07), (-0.02,-1.46)]

Mission Goal: **RIGHT**

Thoughts:

- Notable Objects from Perception: **animal** at (-1.93,7.00)
- Potential Effects from Prediction: **within the safety zone of the ego-vehicle at the 1.0-second timestep**
- Notable Objects from Perception: **adult** at (-1.21,6.78)
- Potential Effects from Prediction: **within the safety zone of the ego-vehicle at the 1.0-second timestep**

Meta Action: **TURN RIGHT WITH A CONSTANT SPEED**

Trajectory:

[(0.11,1.14), (0.45,2.28), (1.12,3.47), (2.18,4.54), (3.65,5.29), (5.49,5.58)]



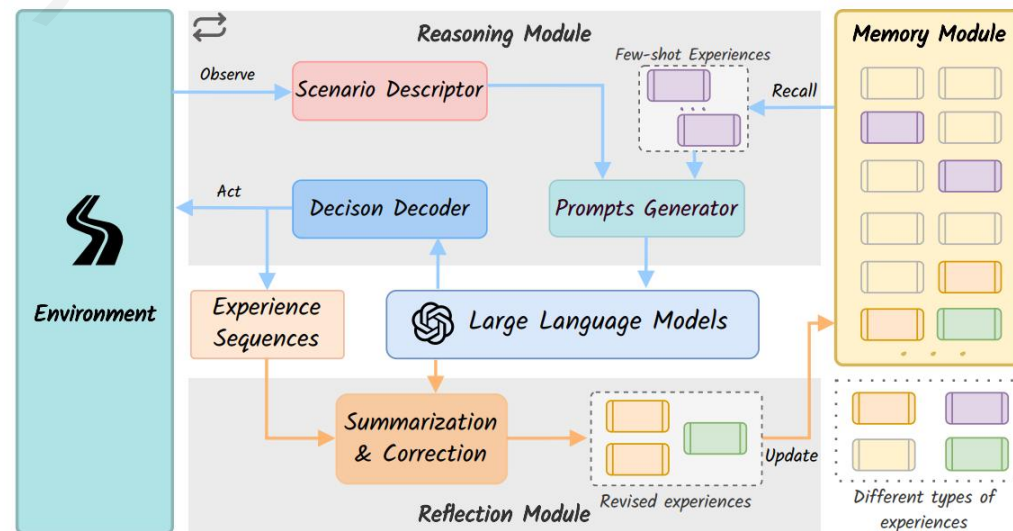
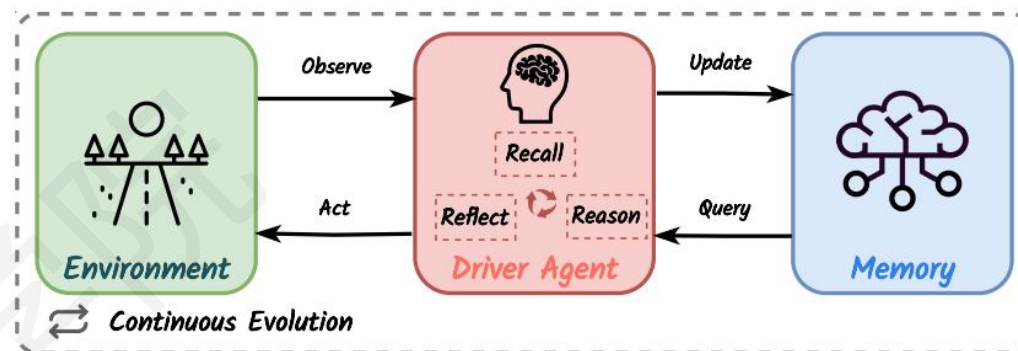


基于场景文本特征的规划模型



□ DiLu

- 自动驾驶系统的知识驱动范式
 - 环境：交互对象
 - 智能体：基于大模型的规划器
 - 记忆：用于存储驾驶经历
- 智能体利用大模型和存储模块
 - 推理：基于常识进行决策
 - 回忆：能够从驾驶经历中获取经验
 - 反思：总结修正经验教训，并存储到记忆模块





基于场景文本特征的规划模型



□ DiLu

• 记忆模块

向量数据库。Key是场景的语义向量化表征，而Value则以自然语言文本形式保存了这个场景历史决策。

• 推理模块

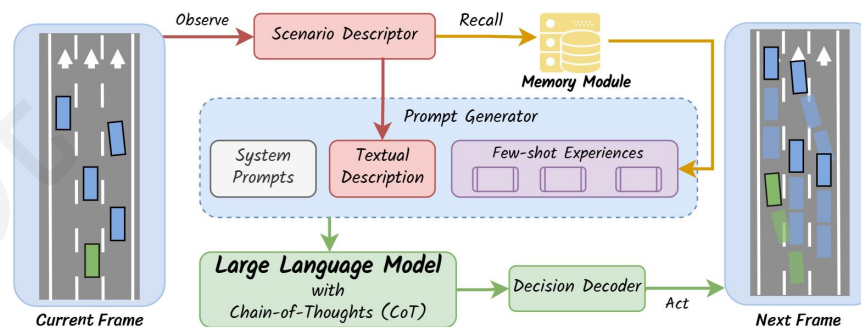
综合common sense与zero-shot推理。对环境进行编码，利用编码从记忆模块中query相似场景。将这个信息作为prompt，和当前场景一起输入到大模型中。

• 反思模块

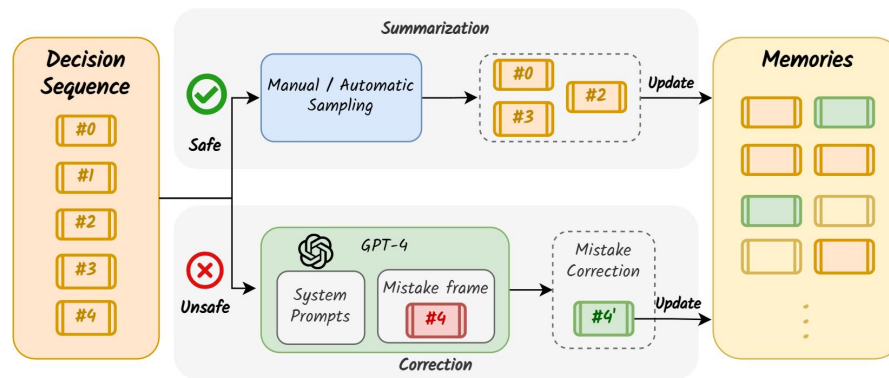
决策正确则总结：将此时场景表征向量作为key，决策说明作为value来更新记忆模块中的经验。

决策错误则矫正：如发生了碰撞或者其他危险行为，借助大模型进行自我反思，并将整个反思过程一起加入到记忆模块中。

Reasoning module



Reflection module

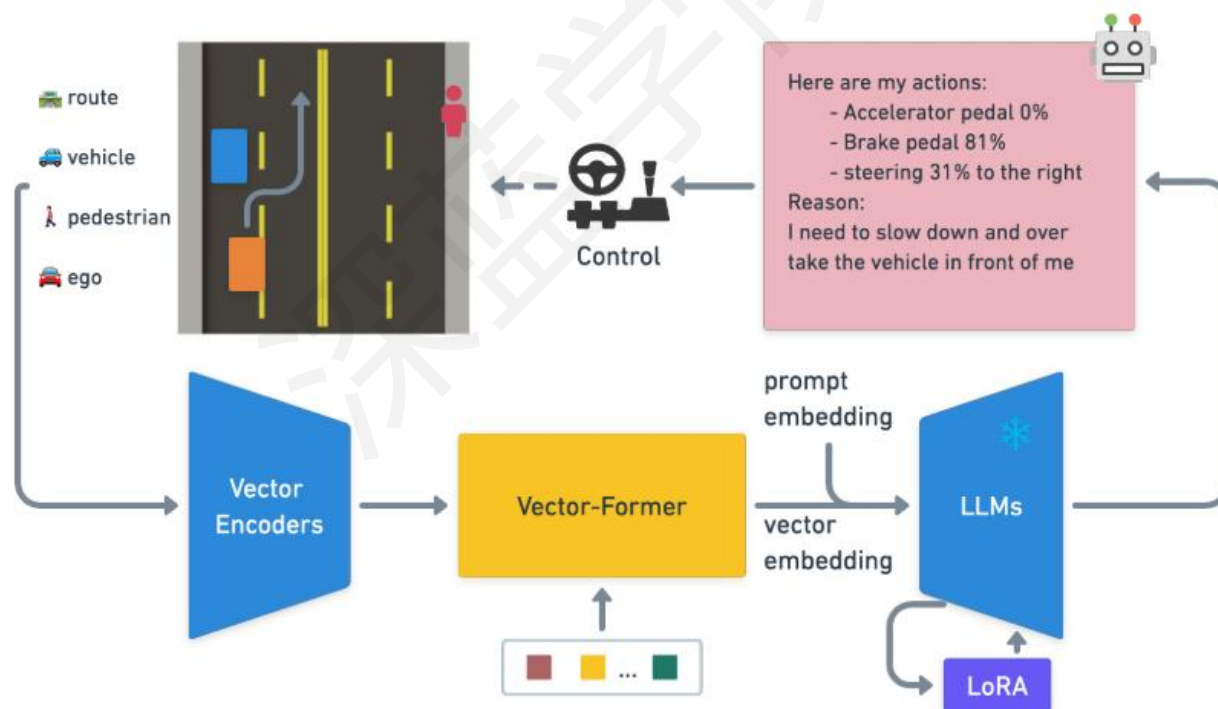




场景特定特征对齐的规划模型

□ Driving with LLMs: Fusing Object-Level Vector Modality for Explainable Autonomous Driving

- How to encode driving scenarios ?
 - 场景结构化编码, 高效表征
- How LLM understands driving scenarios ?





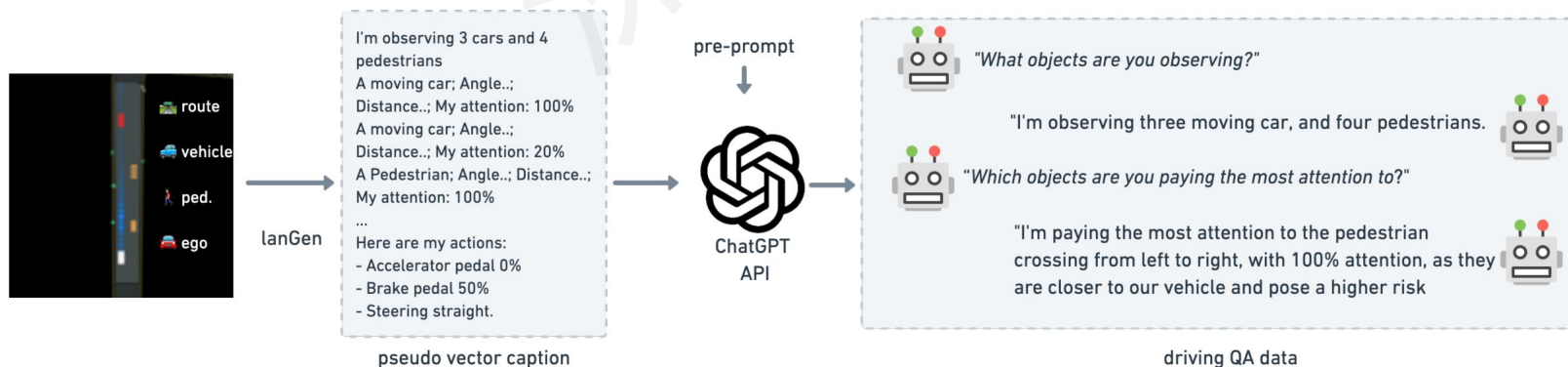
场景特定特征对齐的规划模型

Driving with LLMs: Fusing Object-Level Vector Modality for Explainable Autonomous Driving

- 生成结构化文本数据
- GPT模型构造QA数据

$\text{lanGen}(v_{\text{car}}, v_{\text{ped}}, v_{\text{ego}}, v_{\text{route}}, [o_{rl}]) \rightarrow$

“
A moving car; Angle in degrees: 1.19; Distance: 9.98m; [My attention: 78%]
A pedestrian; Angle in degrees: -41.90; Distance: 11.94m; [My attention: 22%]
My current speed is 11.96 mph.
There is a traffic light and it is red. It is 12.63m ahead.
The next turn is 58 degrees right in 14.51m.
[Here are my actions:]
[- Accelerator pedal 0%]
[- Brake pedal 80%]
[- Steering straight]
”

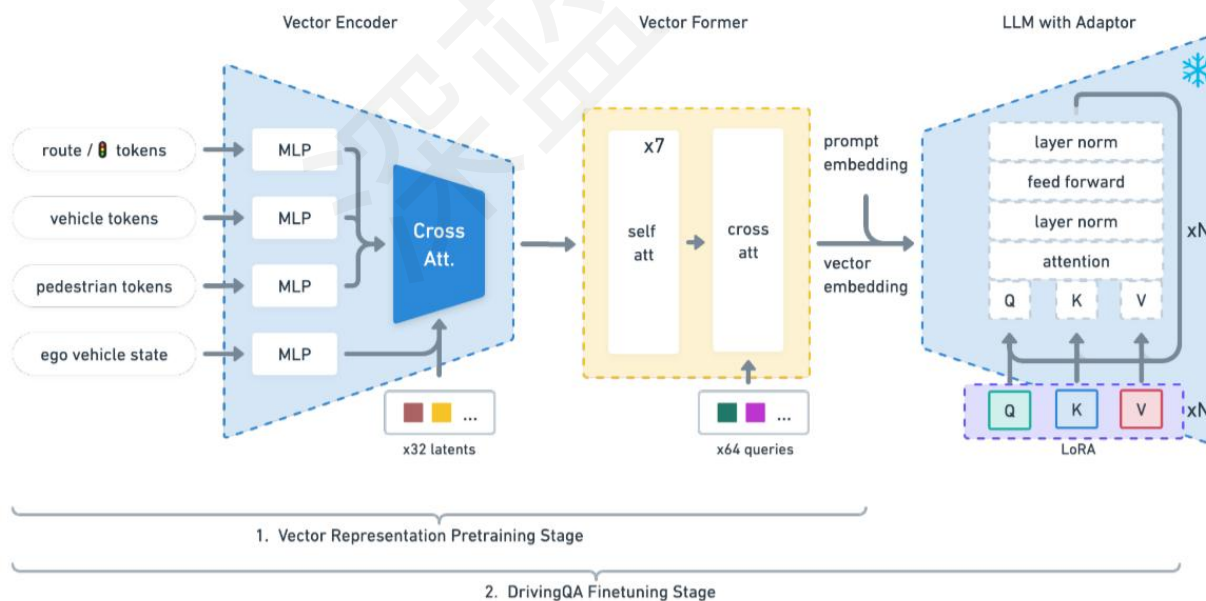




场景特定特征对齐的规划模型

□ Driving with LLMs: Fusing Object-Level Vector Modality for Explainable Autonomous Driving

- Stage 1: Vector Representation Pre-training
 - 冻结LLM, 训练encoder和projector
 - 使用**场景描述数据**监督, 使得LLM能够理解场景信息
- Stage 2: Driving QA Finetuning
 - 冻结encoder和projector, 微调LLM
 - 使用**问答推理数据**微调LLM, 使得LLM具备驾驶场景下推理能力





场景特定特征对齐的规划模型

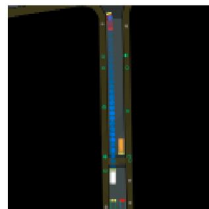


Driving with LLMs: Fusing Object-Level Vector Modality for Explainable Autonomous Driving



How are you going to drive now? How are you going to drive in 10 seconds?

I'm going to maintain my current speed and direction. In 10 seconds, I'm going to turn right.



how far is the closest car? are you going to crash into it? why?

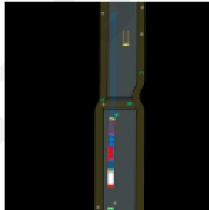
The closest car is 12.18m away from us. We are not going to crash into it because it is driving in the opposite direction.



What's your current speed? can you drive at 40mph in this scenario?

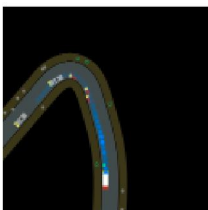
My current speed is 0.00 mph. I can drive at 40 mph in this scenario.

[fail to reason about speed limit and traffic complexity]



how many cars can you see, what are their distances?

I'm observing 2 cars. The first one is at -0.36 degrees and 8.91m away. The second one is at -0.43 degrees and 18.44m away.</s>



What can you observe? what is the closest agent?

I'm observing 1 cars and 3 pedestrians. The closest car is at -11.06 degrees and 25.84m away. The closest pedestrian is at -19.40 degrees and 5.32m away.</s>



What is your current speed, and why do you drive at your current speed?

My current speed is 0.10 mph. I'm driving at this speed because there are no traffic lights and I'm approaching a turn.

[fail to give the actual reason for stopping]

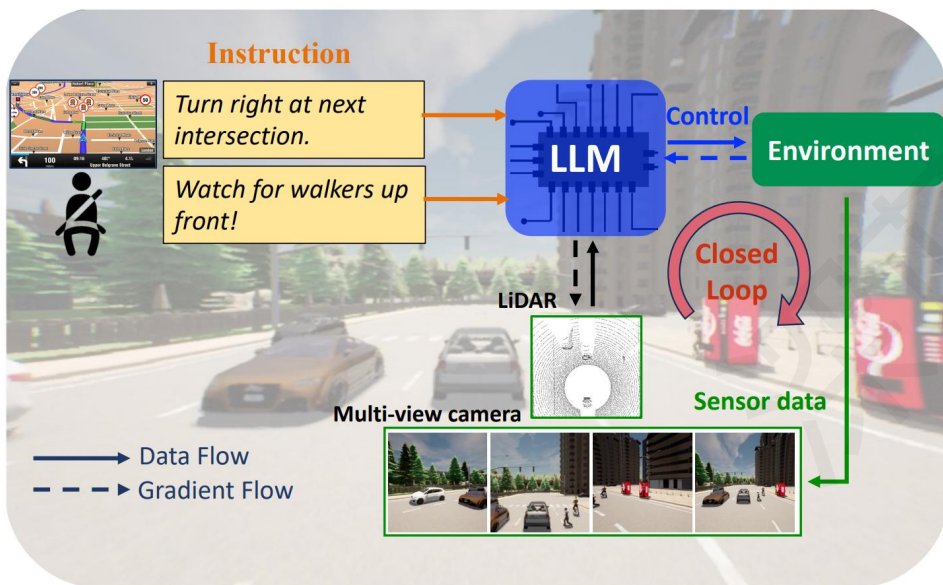




闭环多模态大模型

□ LMDrive

- 使用多模态大模型直接构建闭环端到端自动驾驶系统
 - 处理传感器数据，直接生成车辆控制信号
 - 理解自然语言中的驾驶指令



Navigation instruction: Just move to the left and get ready to leave the highway.



Navigation instruction: Turn left at the next T-junction

Notice instruction: Please watch out for the pedestrians up ahead.



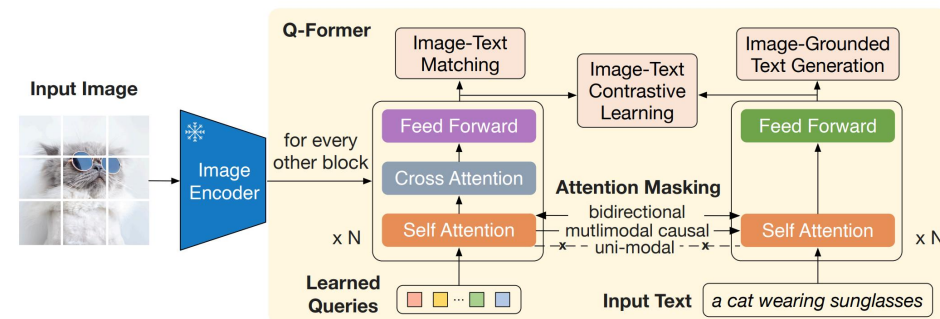
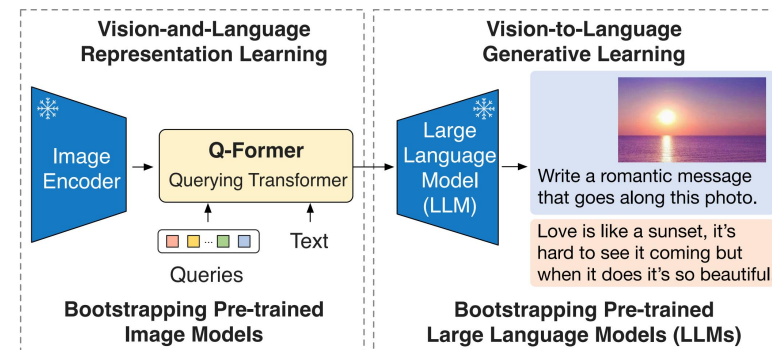
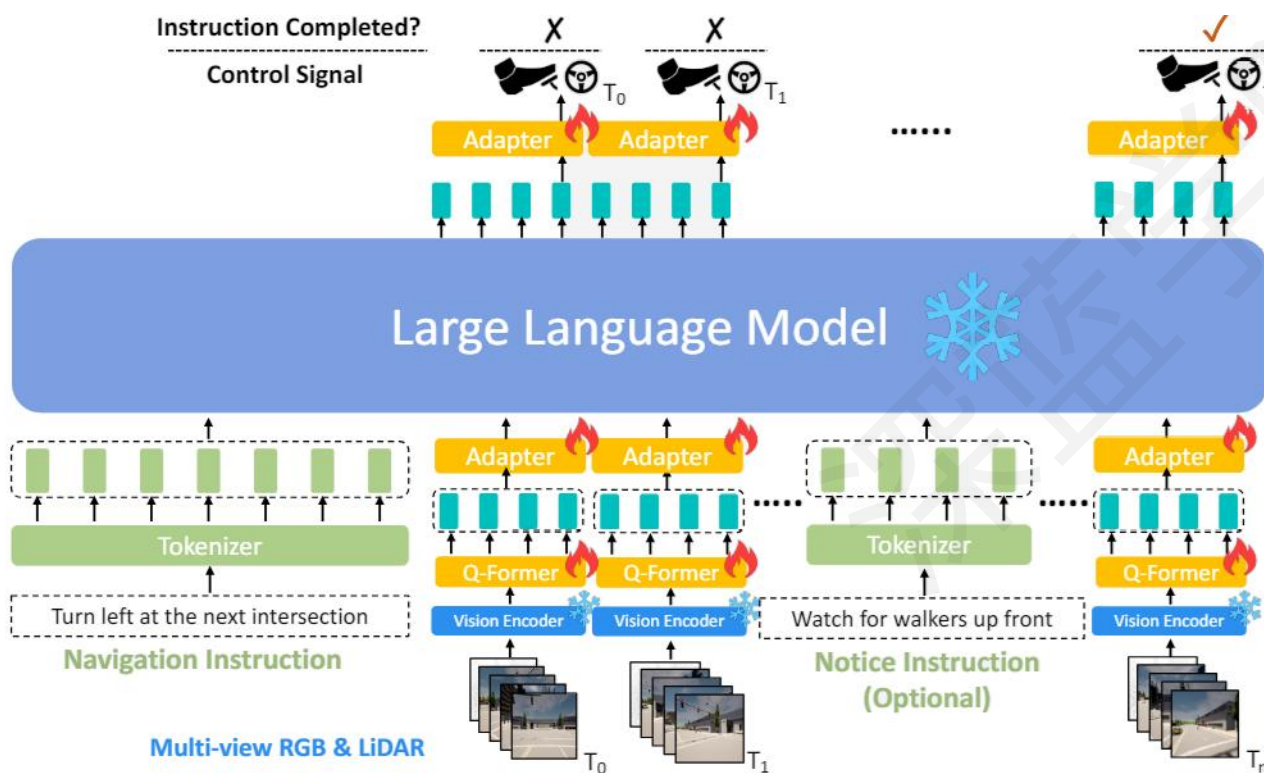


闭环多模态大模型



□ LMDrive

- 以大语言模型为基座，通过Q-Former、Adapter将感知模态输入数据、轨迹模态输出数据align到大模型上



Shao H, Hu Y, Wang L, et al. Lmdrive: Closed-loop end-to-end driving with large language models[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024: 15120-15130.

Li J, Li D, Savarese S, et al. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models[C]//International conference on machine learning. PMLR, 2023: 19730-19742.





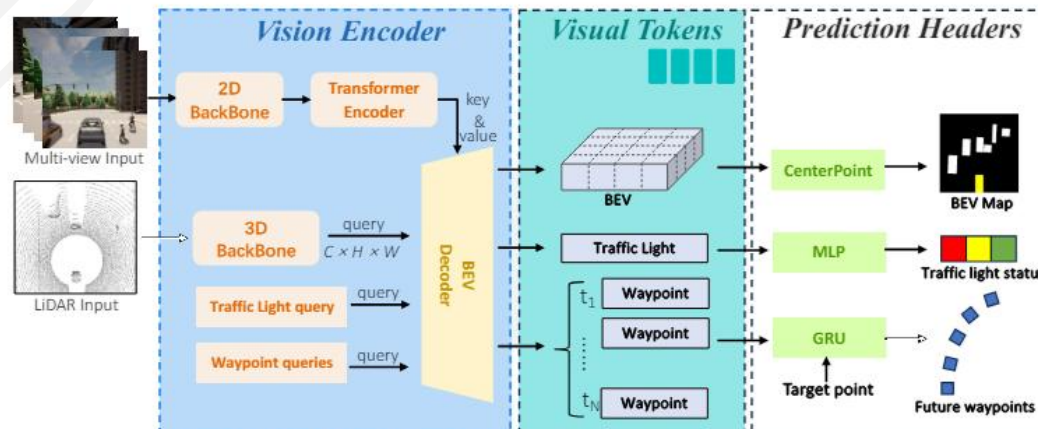
闭环多模态大模型



□ LMDrive 传感器数据处理

- Encoder
 - ResNet提取图像特征图，并使用 k 层transformer融合不同视角的token
 - PointPillars将原始点云数据转换为自车中心的LiDAR特征，并通过PointNet将特征聚合与降采样
- Decoder
 - 使用点云特征与额外的 $N + 1$ 个可学习的query用以生成包含BEV、 n 个waypoint、1个交通灯的视觉token
 - 通过上述处理，将传感器数据转换为适合LLM处理的视觉token，使LMDrive能够通过视觉信息理解和执行驾驶指令

- Pretrain
 - 设置目标检测、未来轨迹waypoint预测、交通灯状态分类三个预训练任务
 - 三个任务仅用于预训练，训练LLM时会丢弃这三个预测头
 - 通过预训练任务，提升vision-encoder的目标检测、路标点预测和交通灯状态分类能力，从而为LMDrive提供更精确和可靠的输入特征。





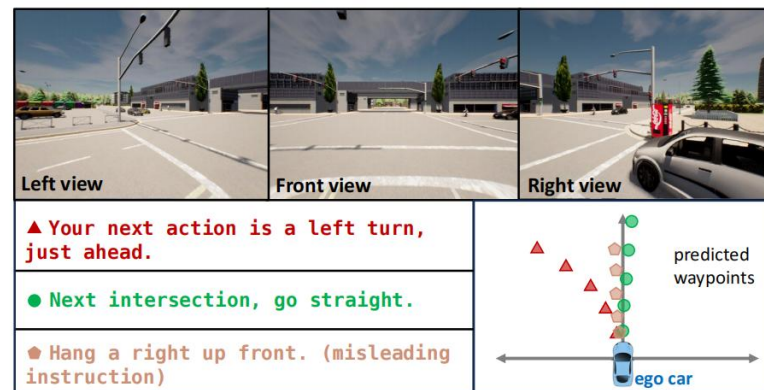
闭环多模态大模型



□ LMDrive 指令跟随自动驾驶LLM

- 训练数据生成：
 - 在CARLA中收集传感器数据和控制信号
 - 通过指令解析和标注收集的数据
 - 标注阶段，将收集到的帧切分为片段，每个片段对应一个导航指令，具体而言，若仿真环境中自车于 T_0 帧开始左转，并在 T_n 帧结束，则 (T_0, T_n) 为一个片段，在此期间如果出现**对抗事件**，则增加一个Notice指令
- 指令token与视觉token：
 - 指令通过LLaMA tokenizer转换为文本token
 - 执行指令的时间从几秒到几分钟不等，在每一帧中，使用所有的历史传感器信息（最大数量为 T_{max} ）来抑制累计误差并改善模型的时间一致性。
 - 由于历史帧的累计会导致视觉token数量快速增长，采用Q-Former通过设置M个可学习的query，将每帧的视觉token减少至M个，随后使用2层MLP将视觉token的维度与文本token进行对齐。
- 动作预测
 - 通过指令与视觉token预测动作token，并通过两层MLP预测未来轨迹的waypoint，并标记指令是否完成。

Type	Three randomly chosen instructions of each instruction type
Follow	Maintain your current course until the upcoming intersection.
	In [x] meters, switch to left lane.
	Ease on to the left and get set to join the highway.
Turn	After [x] meters, take a left.
	At the next intersection, just keep heading straight, no turn.
	You'll be turning left at the next T-junction, alright?
Others	Feel free to start driving.
	Slow down now.
	Head to the point, next one's [x] meters ahead, [y] meters left/right.
Notice	Watch for walkers up front.
	Just a heads up, there's a bike ahead.
	Please be aware of the red traffic signal directly in front of you.





闭环多模态大模型



□ LMDrive LangAuto基准测试

- 特点：
 - 不使用离散的驾驶命令或目标路标来引导代理车辆，仅提供自然语言的导航指令和可选的注意指令
- 范围：
 - 覆盖了CARLA中所有8个公开可用的城镇、16种环境条件的组合（包括7种天气条件和3种白昼条件）
- 测试：
 - LangAuto：对于每条路线，根据代理车辆的当前位置提供并更新导航指令，并根据路线长度分为三个不同的区间。
 - LangAuto-Notice：在LangAuto的基础上，额外添加通知指令给代理车辆。模拟乘客或其他辅助系统在长尾或复杂环境实时提供指令。
 - LangAuto-Sequential：在LangAuto的基础上，将10%的连续2到3个指令合并为一个单一的长指令。
- 指标：
 - 路线完成率(RC)：已完成的总路线长度的百分比
 - 违规得分(IS)：发生碰撞或交通规则违规时，违规得分会按照相应的折扣因子进行衰减
 - 驾驶得分(DS)：路线完成率和违规得分的乘积，描述了驾驶的进展和安全性（通常来讲的主要排名指标）



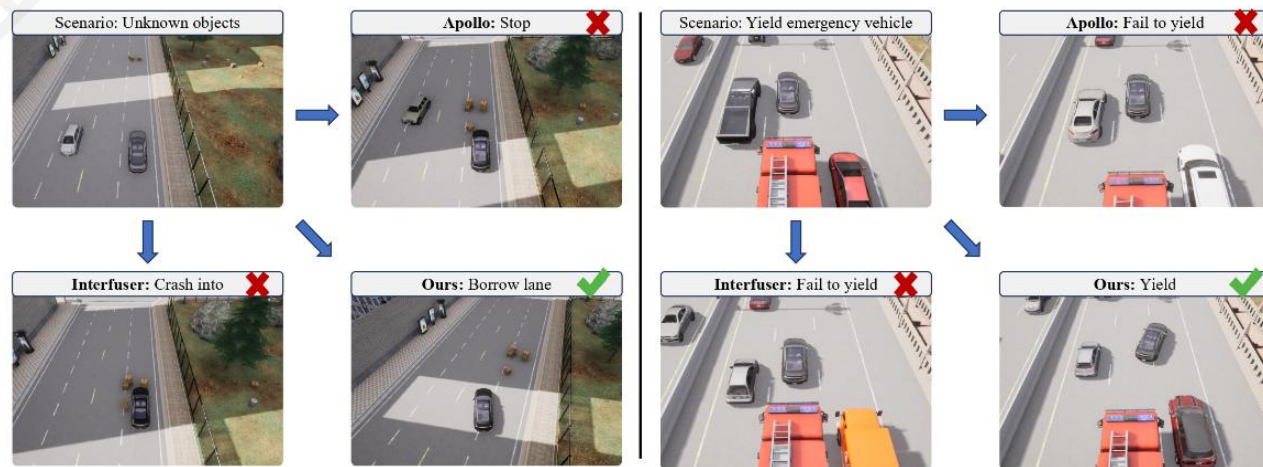
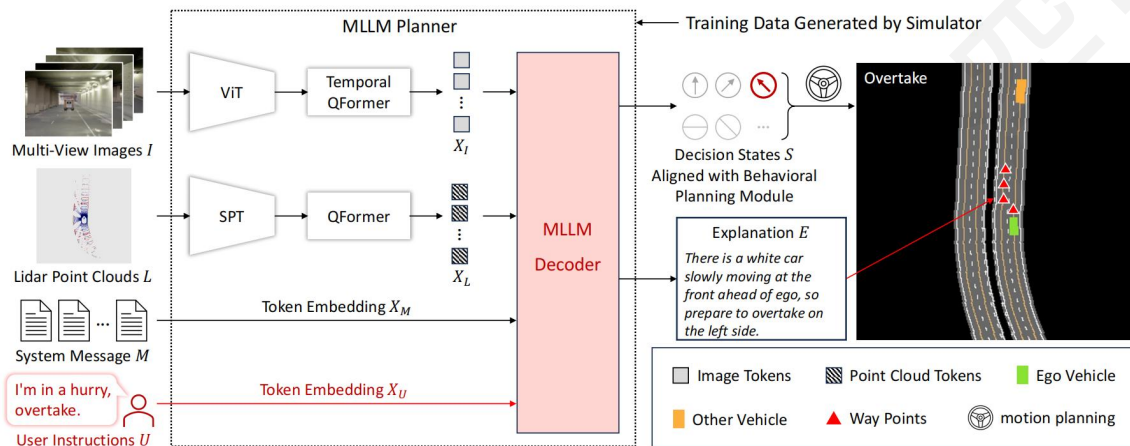


闭环多模态大模型



DriveMLM 将大型语言模型 (LLMs) 的语言选择转化为可执行的控制信号

- 将决策过程分为两类：速度决策和路径决策
 - 速度决策状态包括[保持当前速度 (KEEP), 加速 (ACCELERATE), 减速 (DECELERATE), 停止 (STOP)]
 - 路径决策状态包括[跟随 (FOLLOW), 左变道 (LEFT CHANGE), 右变道 (RIGHT CHANGE), 左借道 (LEFT BORROW), 右借道 (RIGHT BORROW)]
- 将输出分为决策状态和决策解释
- 监督方法：通过交叉熵损失与下一个token预测相结合



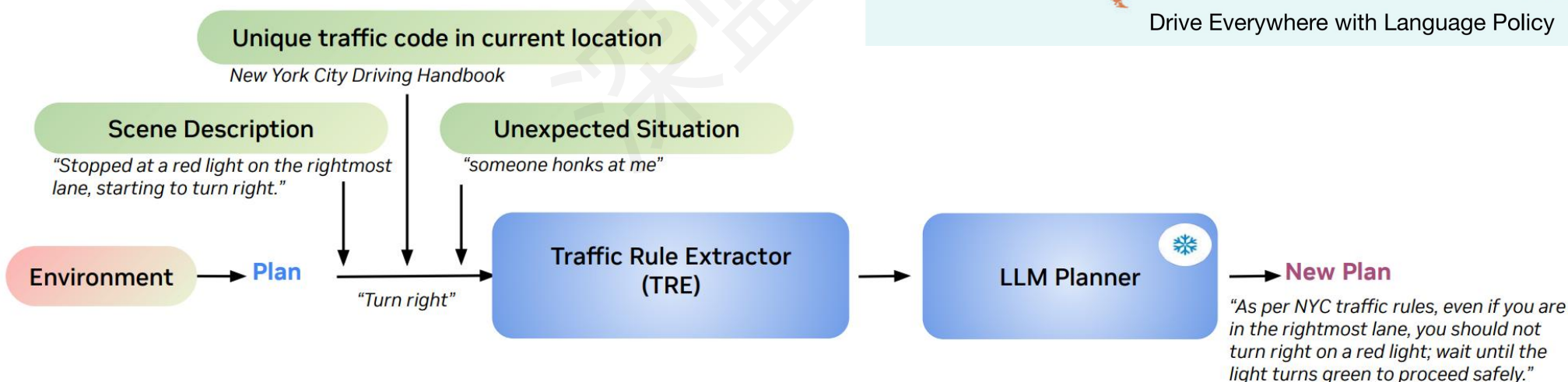


大模型特性探索



Generalization: LLaDA

- 利用大模型，使驾驶行为适应新环境、新规则和新法律
 - 输入：初始规划；当地特定交通法规；极端场景描述
 - 输出：适应后的新规划结果
- 应用场景：
 - 为游客提供交通规则辅助
 - 自动驾驶系统运动规划自适应调整





大模型特性探索



Agents: AgentDriver

- 在nusenes端到端开环任务上，利用大模型的tool using、memory、reasoning能力构建自动驾驶系统

Tool Library

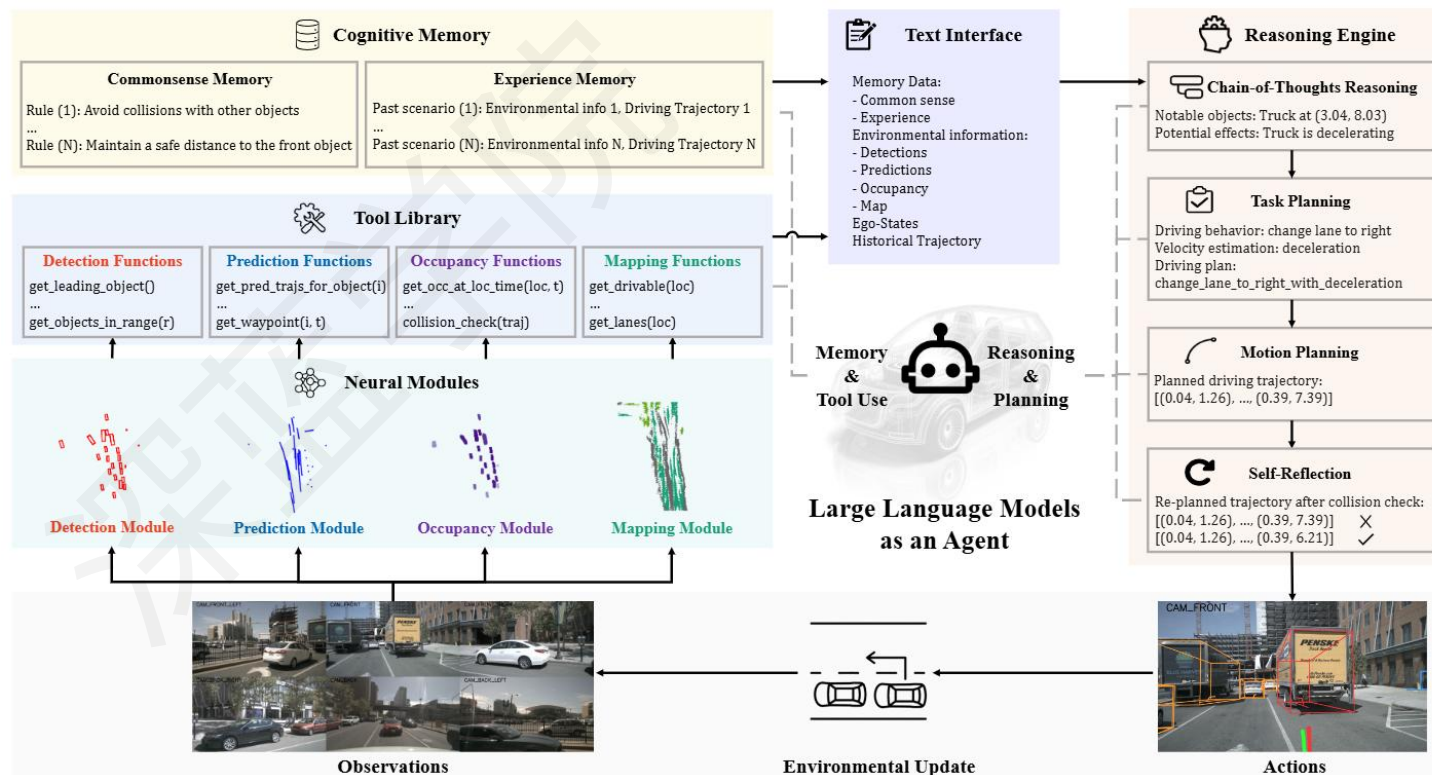
- 网络提供感知预测规划信息
- Agent按照需求调用API

Cognitive Memory

- 常识记忆库（交通规则等）
- 经验记忆库（历史帧经验）

Reasoning Engine

- 链式思维（关键信息、潜在危险）
- 任务规划（高层行为规划）
- 运动规划（低层轨迹规划）
- 自我反思（碰撞检测、重规划）



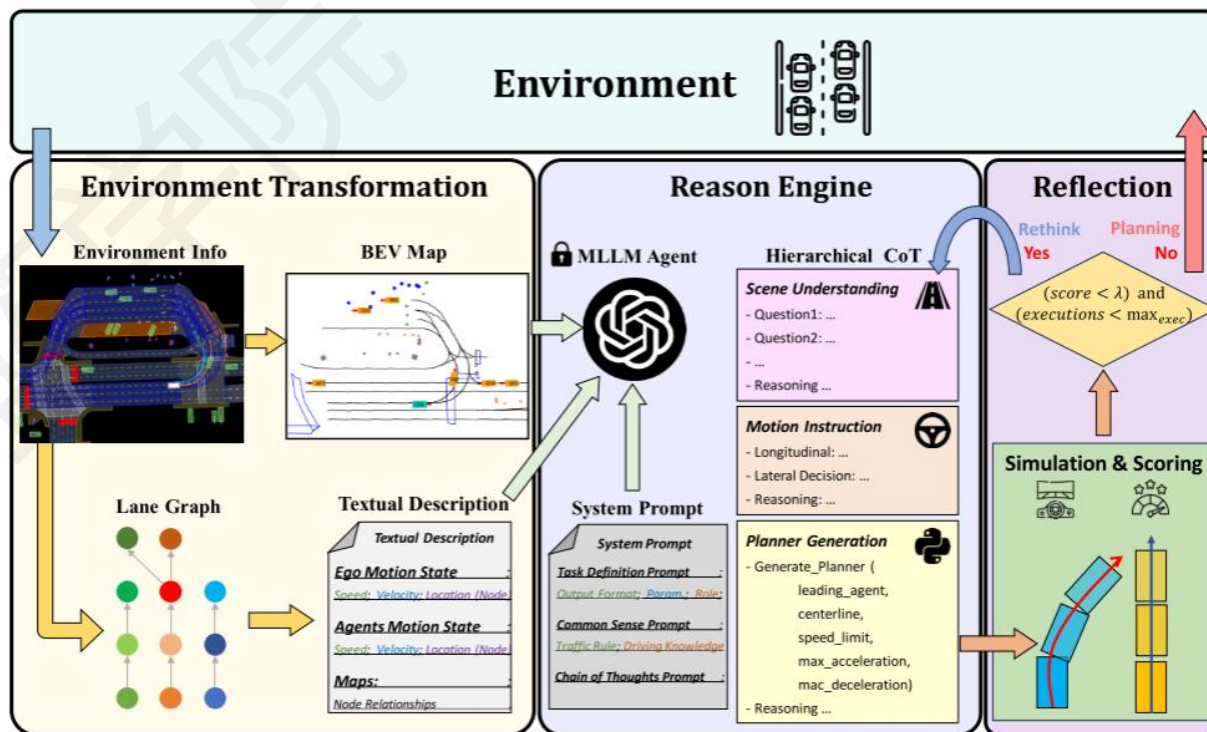


大模型特性探索



Agents: PlanAgent

- 在nuplan闭环规划任务上，利用大模型对场景进行zero-shot推理，生成规划方案
- Environment Transformation
 - BEV Map (全局信息)
 - Lane Graph Text (局部信息)
- Reason Engine
 - System Prompt Formulation
 - System prompts
 - Scenario prompts
 - Hierarchical Chain-of-thought
 - Scene understanding(全局与局部)
 - Motion instruction (横向与纵向)
 - Planner generation (参数化代码)
- Reflection
 - IDM Simulation and Scoring

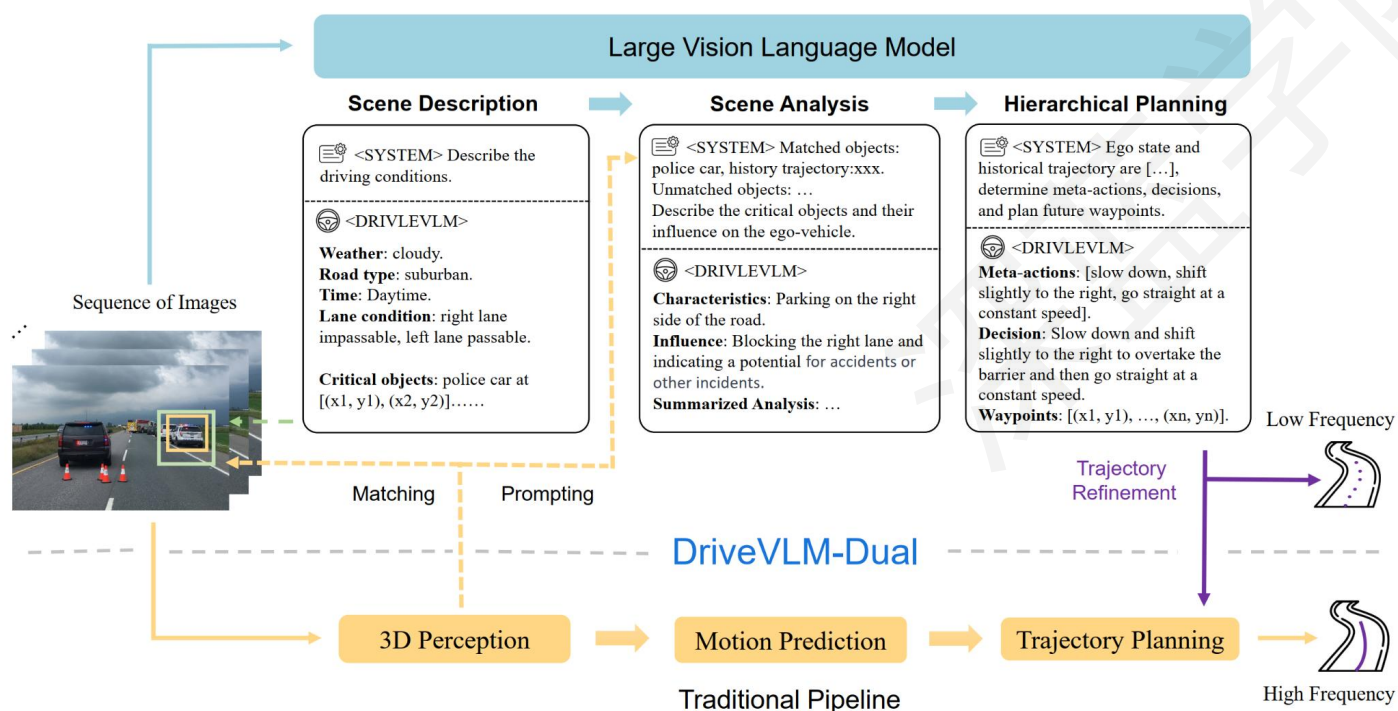




□ Dual System: DriveVLM

- Thinking, Fast and Slow
 - Fast System(90%): Intuitive, Conventional
 - Slow System(10%): Logical, Complex

DriveVLM



- Slow System:
 - VLM based System
 - Scene Description
 - Environment description
 - Critical object identification
 - Scene Analysis
 - Critical Object Analysis
 - Scene-level Summary
 - Hierarchical Planning
 - Meta-actions
 - Decision description
 - Trajectory waypoints
- Fast System:
 - E2E System(UniAD, VAD...)
 - Rule based System(Apollo...)





自动驾驶与多模态大模型



局限与挑战

- 如何进行有效的闭环评测;
- 如何构建有效的长尾场景benchmark, 用于泛化能力验证;
- 如何实现能够实时运行的模型部署;
- 如何有效挖掘多模态大模型的常识推理能力, 并与自动驾驶任务对齐。

